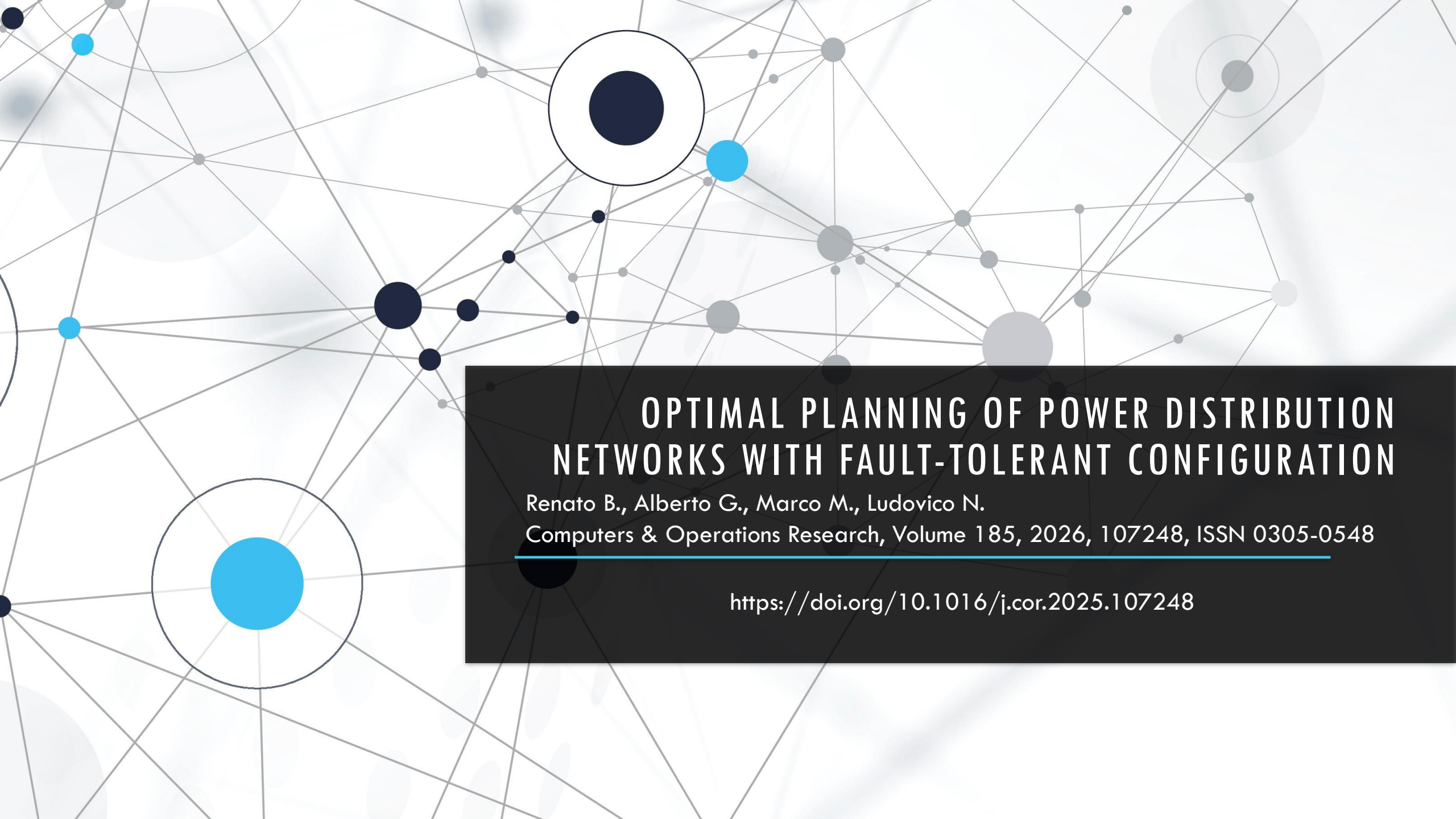


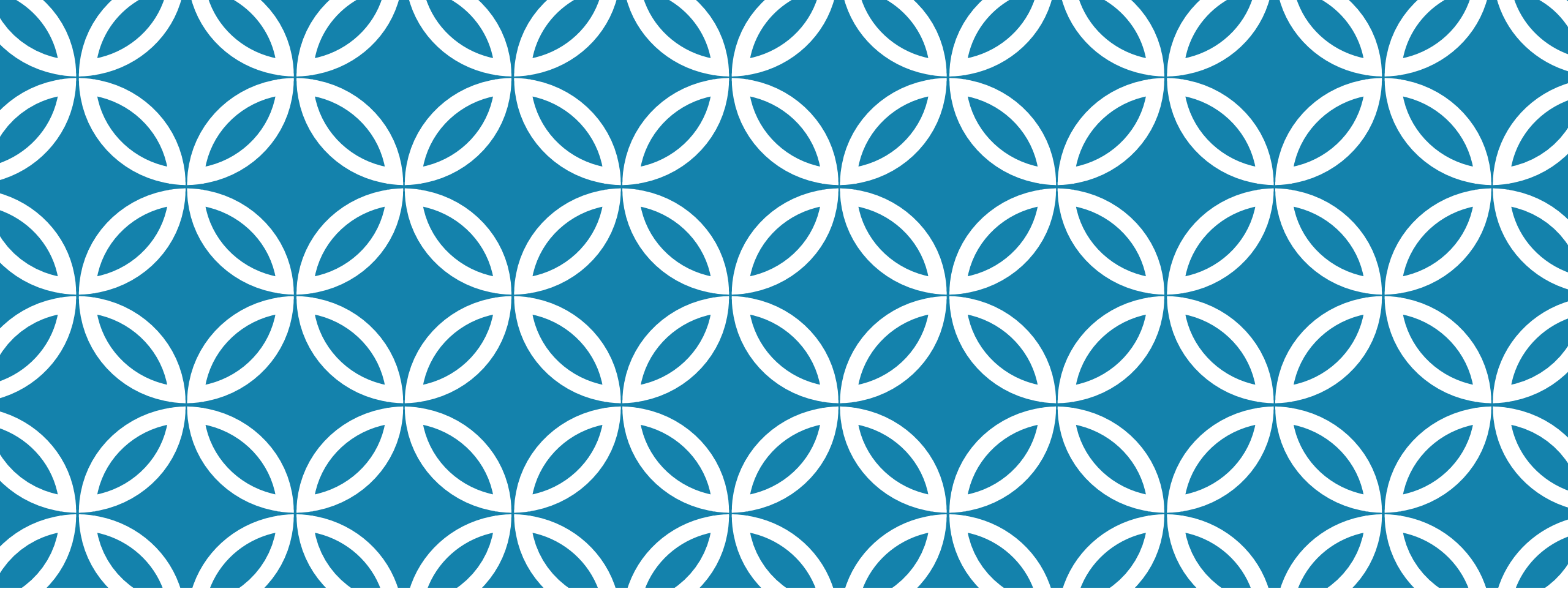
The background of the entire page is an abstract network diagram. It consists of numerous nodes of varying sizes and colors (dark blue, light blue, and grey) connected by thin, light grey lines. Some nodes are highlighted with larger, concentric circles. The overall layout is a complex web of connections, suggesting a network structure.

OPTIMAL PLANNING OF POWER DISTRIBUTION NETWORKS WITH FAULT-TOLERANT CONFIGURATION

Renato B., Alberto G., Marco M., Ludovico N.

Computers & Operations Research, Volume 185, 2026, 107248, ISSN 0305-0548

<https://doi.org/10.1016/j.cor.2025.107248>



CASE STUDY

DISTRIBUTION NETWORKS

A distribution network is a system of electrical components that delivers electricity from the transmission system to the end-users. It consists of:

- **Primary Substations (PSs)** also known as *source nodes*
- **Secondary Substations (SSs)** also known as *demand nodes*



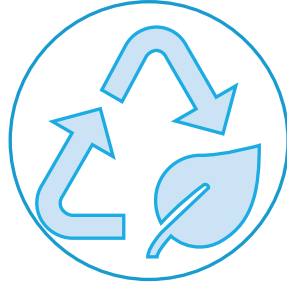
PROBLEMS AND CHALLENGES



Reliable: It should be able to withstand faults and continue to provide electricity to customers.



Cost-effective: It should be designed and operated in a way that minimizes costs while ensuring reliable service.



Sustainable: It should be designed and operated in a way that minimizes environmental impact.



Resilient: It should be able to recover quickly from disruptions and adapt to changing conditions.



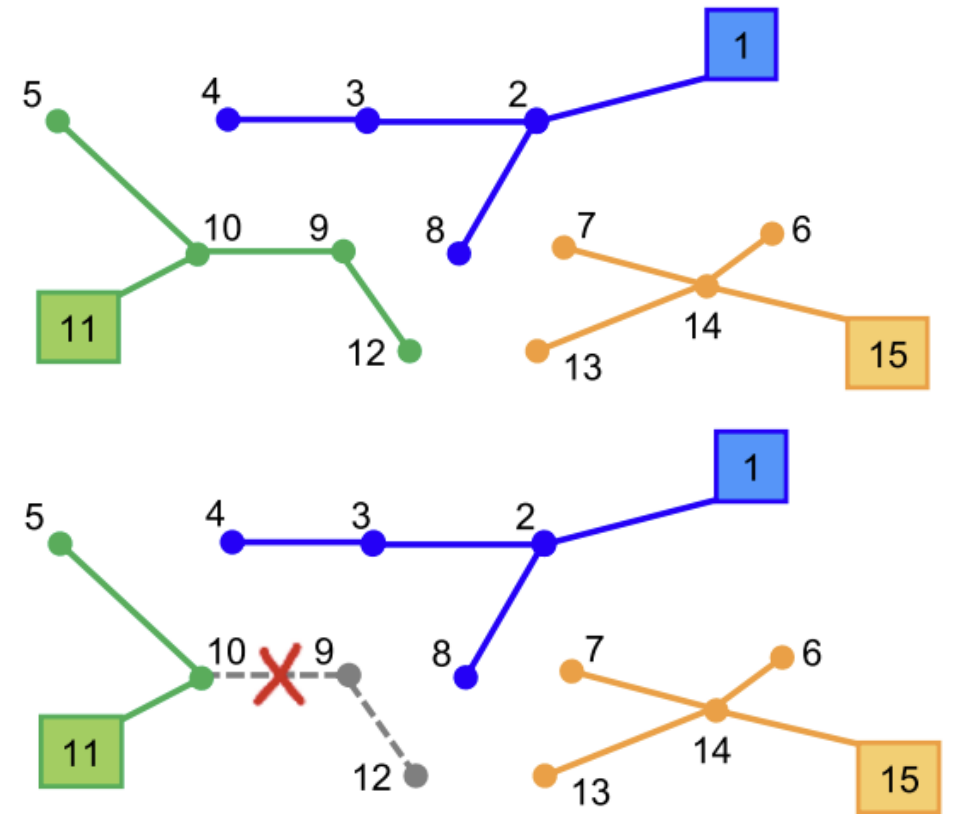
Scalable: It should be able to accommodate growth in demand and the integration of new technologies.



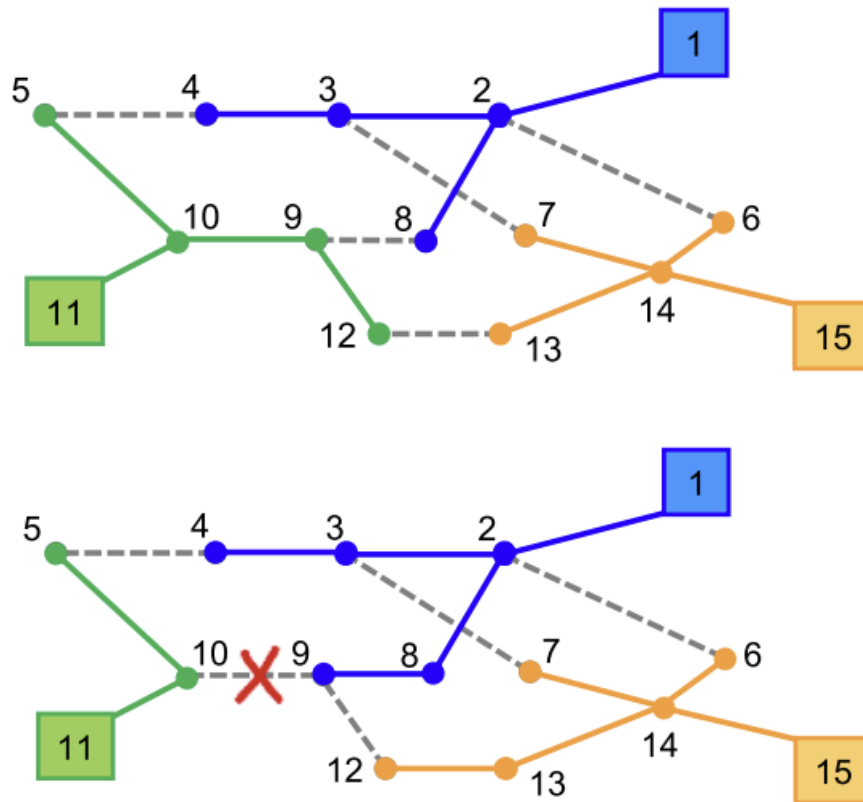
THE ORIGINAL APPROACH

Originally, distribution network planning was based around **radial configurations**, where each demand node is connected to a single source node. This approach is simple and cost-effective, but it has some limitations:

- **Lack of redundancy:** If a fault occurs in a radial configuration, the affected demand nodes will lose power until the fault is repaired.
- **Limited flexibility:** In case of maintenance or upgrades, the affected demand nodes will lose power until the work is completed.



MESHED NETWORK TOPOLOGIES



To address the limitations of radial configurations, **meshed networks** add redundant arcs that form cycles between substations:

- **Alternative paths:** unused arcs reroute power when a branch fails or requires maintenance.
- **Fast reconfiguration:** the DSO can quickly restore power to affected demand nodes, minimising downtime.
- **Improved reliability:** fault tolerance achieved through topology alone, without explicit fault simulation.

CONTENT OF THE WORK

The key research gap being addressed is that prior works relying on explicit fault scenarios to ensure network redundancy tend to become computationally intractable at scale.

The proposed models achieve fault tolerance through topology alone, without simulating post-fault reconfiguration scenarios.

Three MILP formulations are introduced:

- Two target **loop-feeder topology** (one using Single Commodity Flow constraints, one using Subtour Elimination Constraints).
- One targets the more demanding **open-loop topology** (also based on Subtour Elimination), with an enhanced variant that adds cut-based constraints to improve performance.

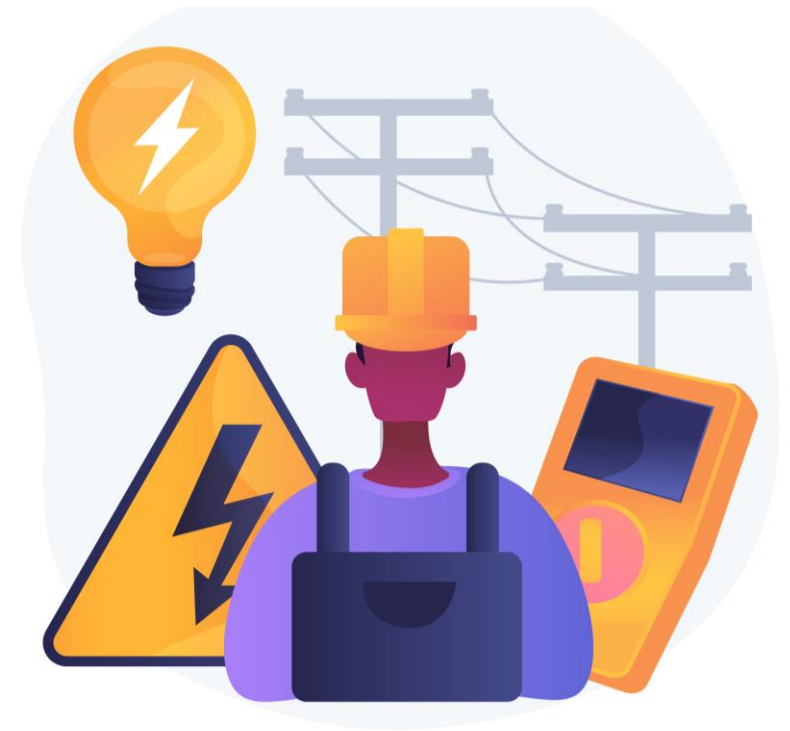
More on this later...

PROBLEM DESCRIPTION

The planning scenario: a DSO builds an entirely new grid from scratch (newly electrified areas, voltage upgrades, or full renovation).

Key assumptions:

- Locations of all **PSs and SSs** are fixed in advance, as is each SS's reference load.
- **No candidate connection pre-restricted:** any substation can be connected to any other.



OBJECTIVES AND CONSTRAINTS

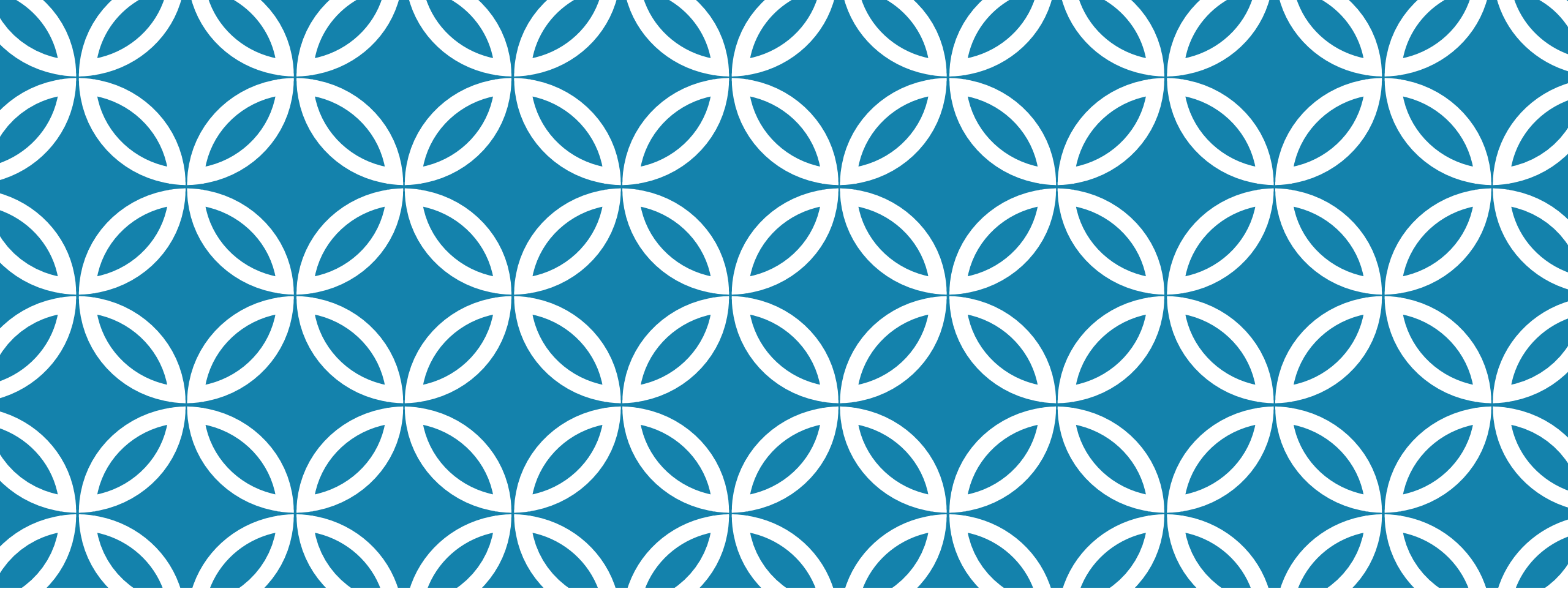
The core objective

Select which physical connections to build so that **total cost is minimized** while guaranteeing two things: Every SS is served and a chosen level of fault tolerance is maintained.

Operational constraints

Electricity must flow radially during normal operation. This is achieved by opening one switch per power line, effectively splitting each line into two feeders, each supplied by a single PS.

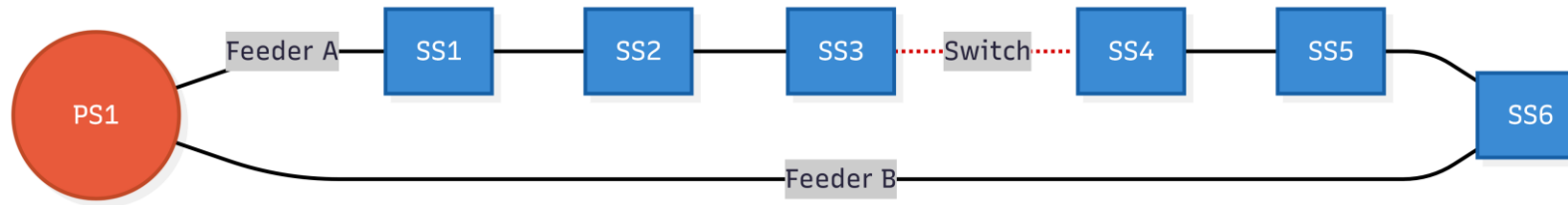
Cable capacity is standardized and deliberately overscaled ensuring that if any single branch fails, the remaining feeder can absorb the full rerouted load without overloading.



PROBLEM FORMULATION



LOOP-FEEDER TOPOLOGY

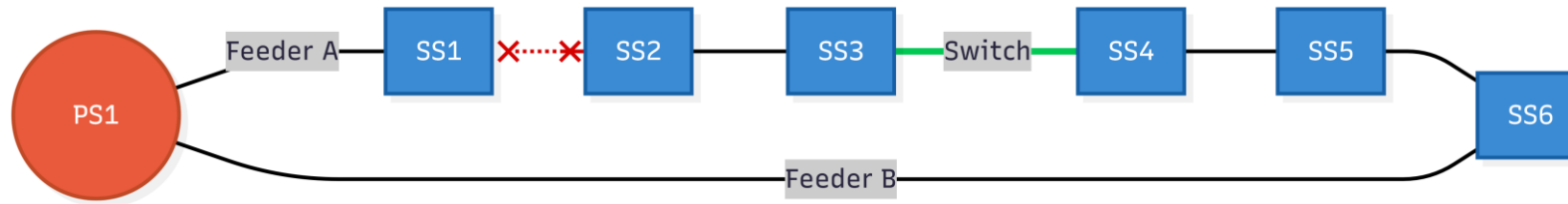


In a loop-feeder topology, each demand node is connected to exactly one source node.

A power line may begin and end with the same source node, forming a loop on a single root. **Resilient to any single branch fault.**

During normal operation one switch is opened to create two radial feeders. If any single branch fails, the other feeder can supply all Secondary Substations (SSs).

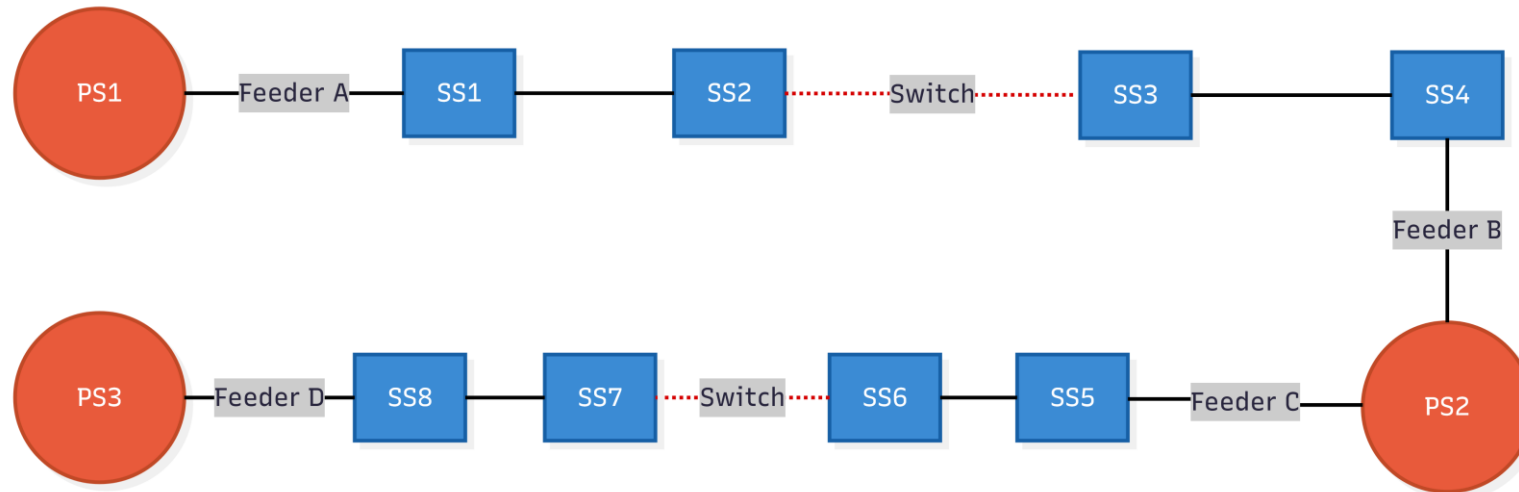
LOOP-FEEDER TOPOLOGY



A fault anywhere on Feeder A or B causes that switch to close, rerouting power from PS1. However, if PS1 itself fails, **all SSs lose supply**.

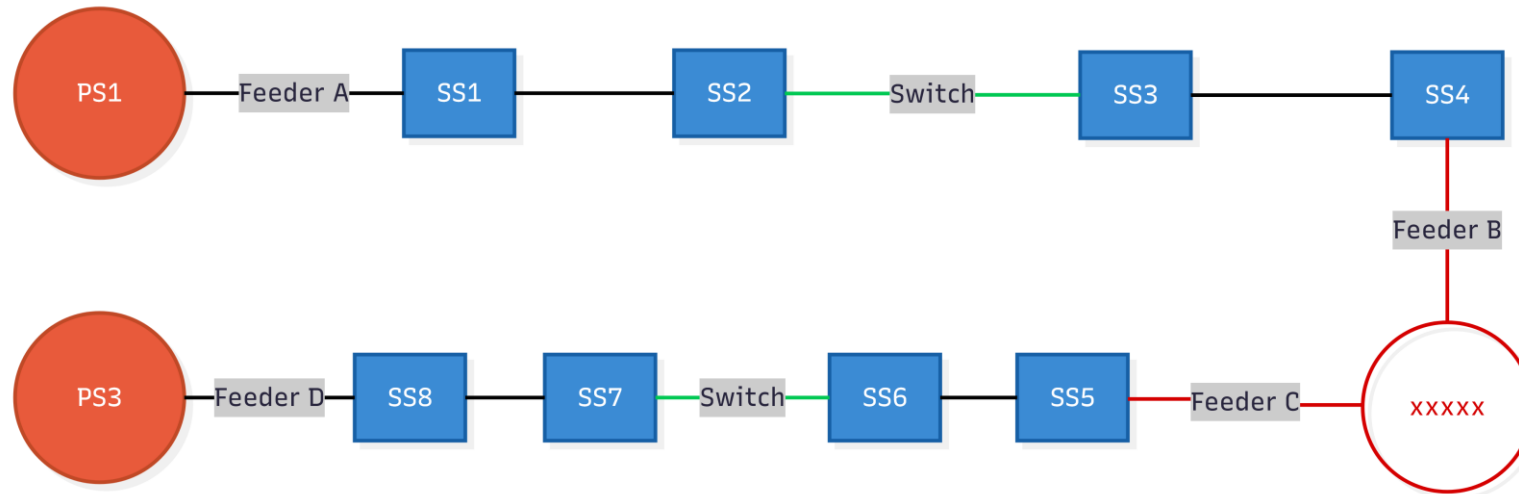
This is the key weakness of loop-feeder.

OPEN-LOOP TOPOLOGY



In this configuration, every power line connects **two different** Primary Substations. This provides stronger resilience: even a full PS failure leaves all SSs reachable via the other PS at the far end of each power line.

OPEN-LOOP TOPOLOGY



If PS2 fails entirely, SS1-SS2 are still served by PS1, and SS5-SS6 fall back to PS3. No SS is left with both ends of its power line attached to the same failed PS.

MIXED INTEGER LINEAR FORMULATION(S)

The model of the distribution network is described as a **directed graph** $G(N, A)$, where nodes represent PSs and SSs , and arcs represent potential electrical connections.

The **fundamental structural requirement** underpinning all three formulations is that every demand node must have exactly **two incident active arcs**, this is what creates the meshed redundancy.

Flow through any active arc is capped at **half its rated capacity**, ensuring that if one branch fails, the surviving feeder can carry the full rerouted load without overloading.

MIXED INTEGER LINEAR FORMULATION(S)

Sets and indices

Description

N	Set of nodes
R	Set of roots, $R \subset N$
D	Set of demand nodes, $D \subset N, D = N \setminus R$
i, j, k	Indices for the nodes
A	Set of arcs (branches)
(i, j)	indices of arcs viewed as ordered pairs of nodes, with $i < j$

MIXED INTEGER LINEAR FORMULATION(S)

Parameters

d_i	Power demand of node i (in MW)
$p_{i,j}$	Power capacity of arc (i,j) (in MW)
$c_{i,j}$	Activation cost of arc (i,j)

Description

Variables

$x_{i,j}$	Activation of arc (i,j) (binary variables)
$f_{i,j}$	Power flowing through arc (i,j) (continuous). If positive the direction is from i to j , if negative from j to i
$s_{i,j}$	Fictitious power flowing through arc (i,j) (continuous). If positive the direction is from i to j , if negative from j to i .

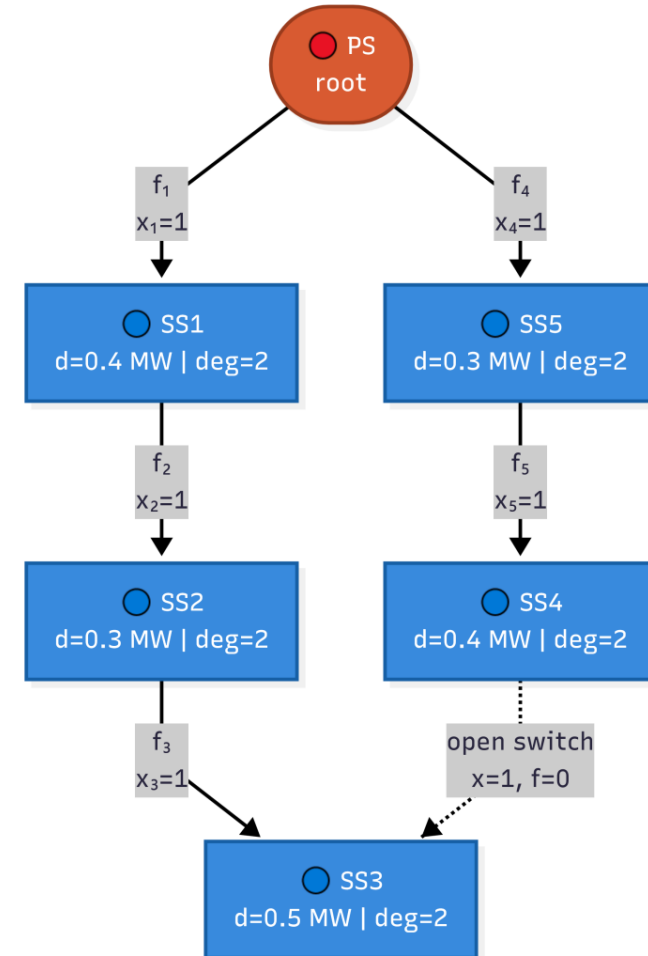
All the variables exists only for $i < j$

LOOP-FEEDER SINGLE COMMODITY FLOW (LF-SCF)

Connectivity is enforced via a **fictitious commodity flow** that must reach every demand node.

Advantage: polynomial number of constraints so it is directly solvable with no constraint generation.

Trade-off: introduces additional continuous flow variables $s(i,j)$ for every arc.



LOOP-FEEDER SINGLE COMMODITY FLOW (LF-SCF)

$$\min \sum_{(i,j) \in A} c_{ij} x_{ij} \quad (1)$$

s.t.

$$\sum_{(i,j) \in \delta(k)} x_{ij} = 2 \quad \forall k \in D \quad (2)$$

$$\sum_{(i,j) \in \delta^-(k)} f_{ij} = d_k + \sum_{(i,j) \in \delta^+(k)} f_{ij} \quad \forall k \in D \quad (3)$$

$$\sum_{(i,j) \in \delta^-(k)} s_{ij} = 1 + \sum_{(i,j) \in \delta^+(k)} s_{ij} \quad \forall k \in D \quad (4)$$

$$-(x_{ij} p_{ij})/2 \leq f_{ij} \leq (x_{ij} p_{ij})/2 \quad \forall (i,j) \in A \quad (5)$$

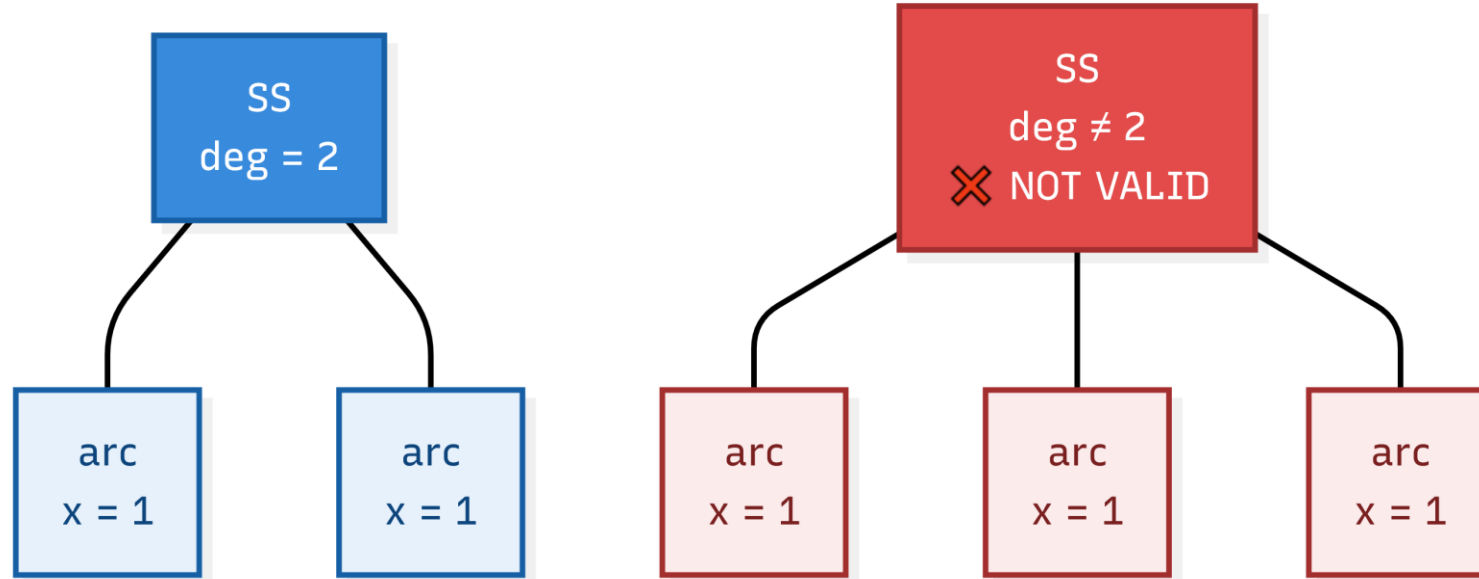
$$-(x_{ij} |D|)/2 \leq s_{ij} \leq (x_{ij} |D|)/2 \quad \forall (i,j) \in A \quad (6)$$

$$x_{ij} \in \{0, 1\} \quad \forall (i,j) \in A$$

LOOP-FEEDER SINGLE COMMODITY FLOW (LF-SCF)

$$\sum_{(i,j) \in \delta(k)} x_{ij} = 2 \quad \forall k \in D \quad (2)$$

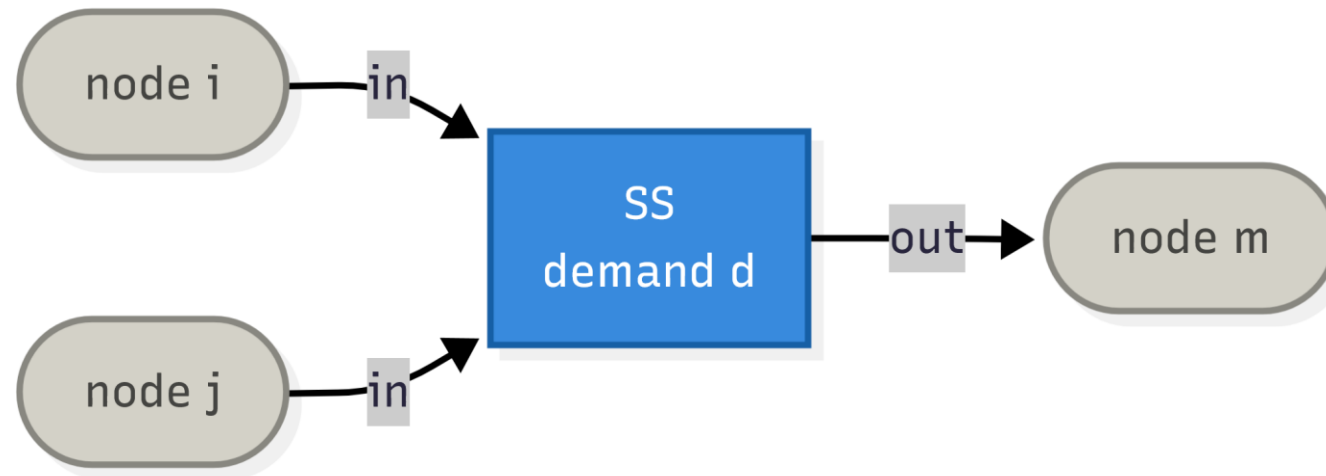
Each demand node has exactly two incident active arcs



LOOP-FEEDER SINGLE COMMODITY FLOW (LF-SCF)

$$\sum_{(i,j) \in \delta^-(k)} f_{ij} = d_k + \sum_{(i,j) \in \delta^+(k)} f_{ij} \quad \forall k \in D \quad (3)$$

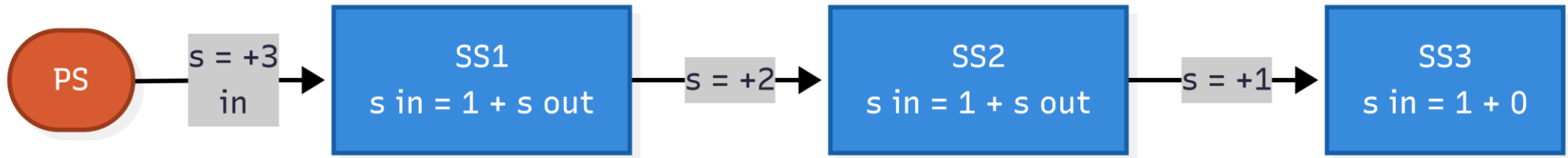
Real power flow conservation: power entering node k equals its demand plus power exiting.



LOOP-FEEDER SINGLE COMMODITY FLOW (LF-SCF)

$$\sum_{(i,j) \in \delta^-(k)} s_{ij} = 1 + \sum_{(i,j) \in \delta^+(k)} s_{ij} \quad \forall k \in D \quad (4)$$

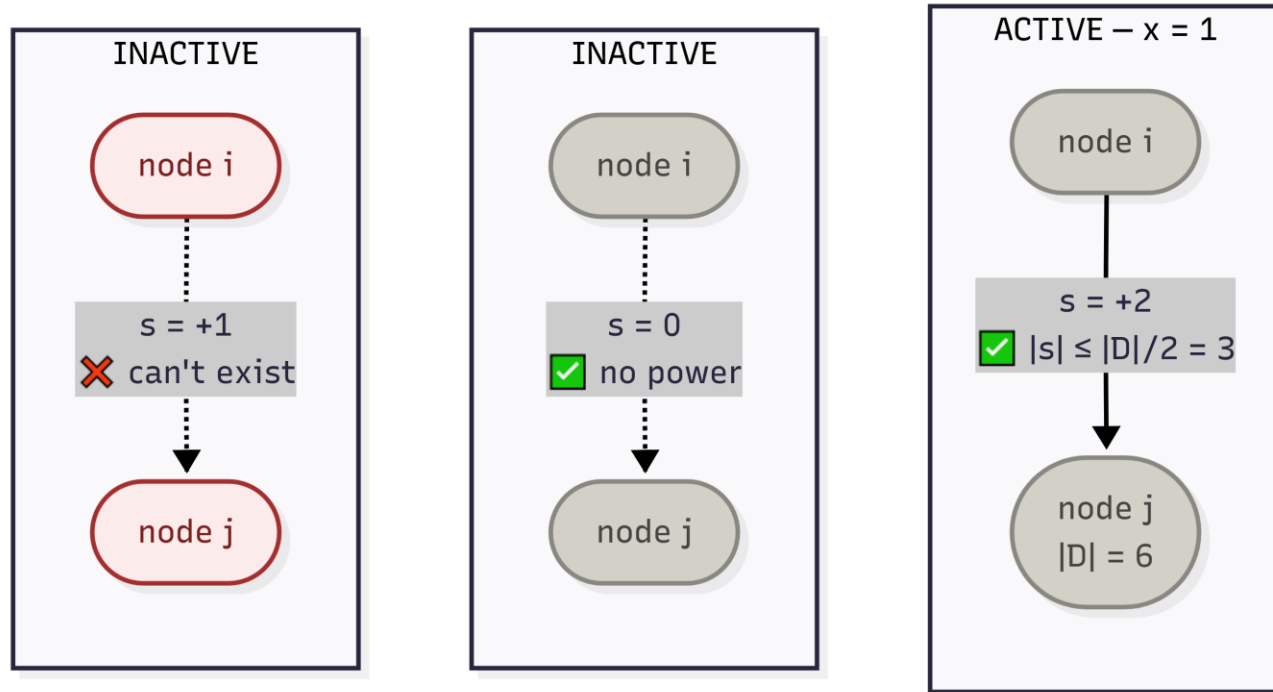
Fictitious flow conservation: the fictitious commodity must reach every node, ensuring no disconnected subtour exists.



LOOP-FEEDER SINGLE COMMODITY FLOW (LF-SCF)

$$-(x_{ij}p_{ij})/2 \leq f_{ij} \leq (x_{ij}p_{ij})/2 \quad \forall (i, j) \in A \quad (5)$$

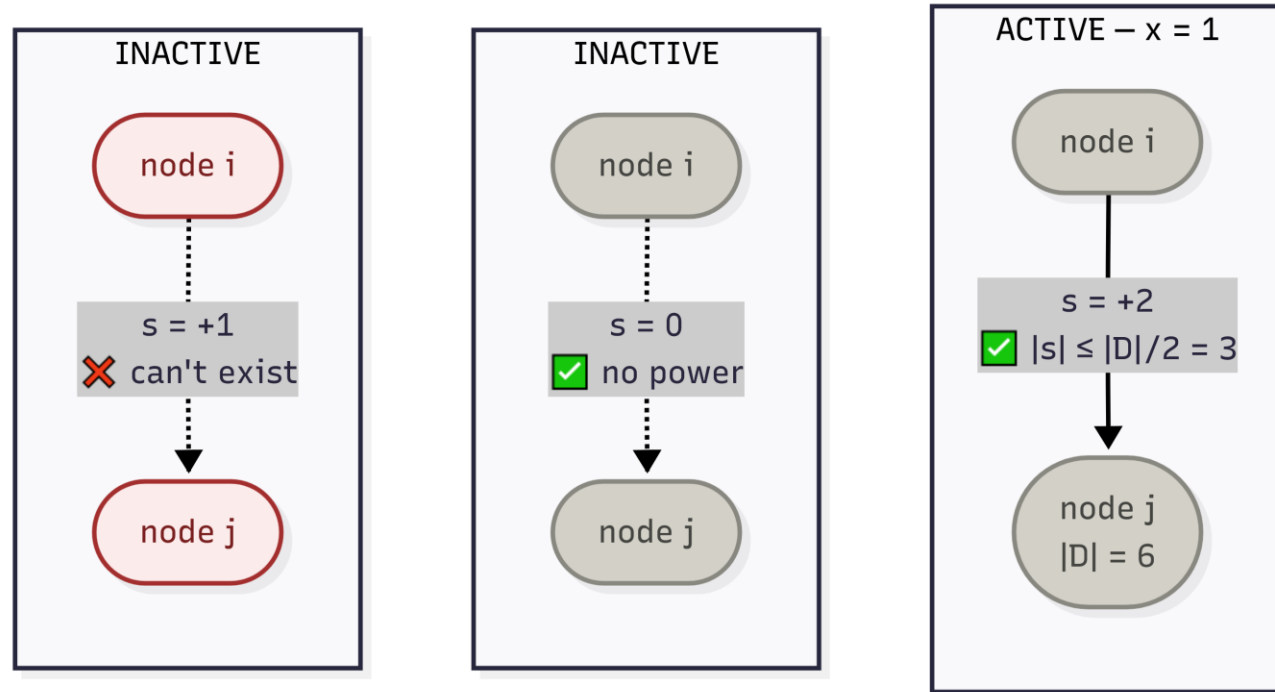
Real power flow is bounded to $\pm$ half capacity if the arc is active, zero otherwise.



LOOP-FEEDER SINGLE COMMODITY FLOW (LF-SCF)

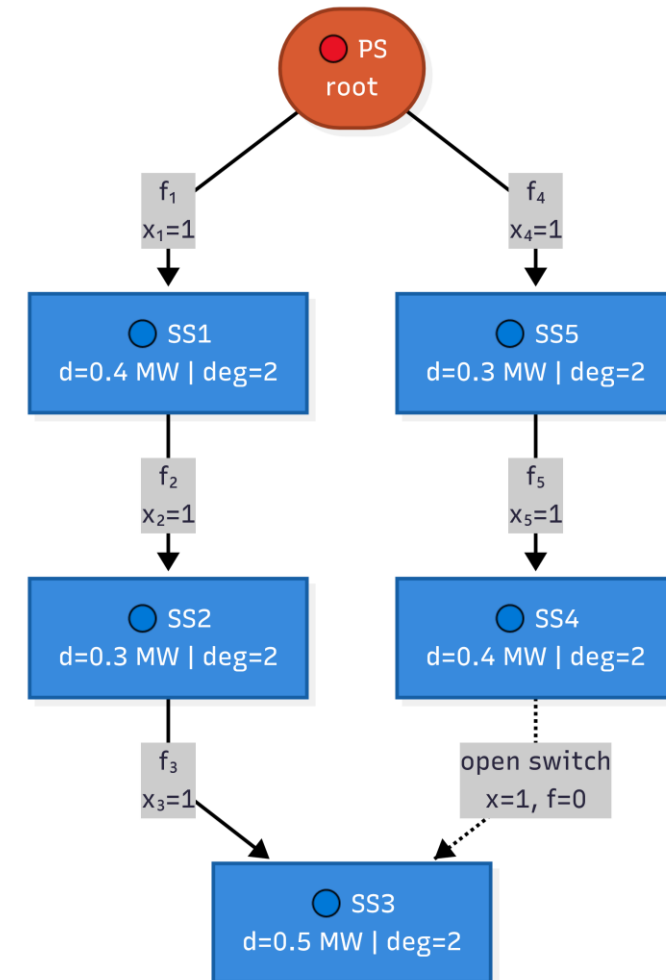
$$-(x_{ij}|D|)/2 \leq s_{ij} \leq (x_{ij}|D|)/2 \quad \forall (i, j) \in A \quad (6)$$

Same logical condition applied to the fictitious commodity flow.



LOOP-FEEDER SUBTOUR ELIMINATION (LF-SE)

This formulation replaces the fictitious commodity flow of LF-SCF with **Subtour Elimination Constraints (SECs)**, avoiding the need for the extra $s(i, j)$ variables entirely. It shares the same objective to minimize total arc activation cost and produces networks with loop-feeder topology.



LOOP-FEEDER SUBTOUR ELIMINATION (LF-SE)

$$\min \sum_{(i,j) \in A} c_{ij} x_{ij} \quad (7)$$

s.t.

$$\sum_{(i,j) \in \delta(k)} x_{ij} = 2 \quad \forall k \in D \quad (8)$$

$$\sum_{(i,j) \in A(S)} x_{ij} \leq |S| - 1 \quad \forall S \subseteq D, |S| \geq 2 \quad (9)$$

$$\sum_{(i,j) \in \delta^-(k)} f_{ij} = d_k + \sum_{(i,j) \in \delta^+(k)} f_{ij} \quad \forall k \in D \quad (10)$$

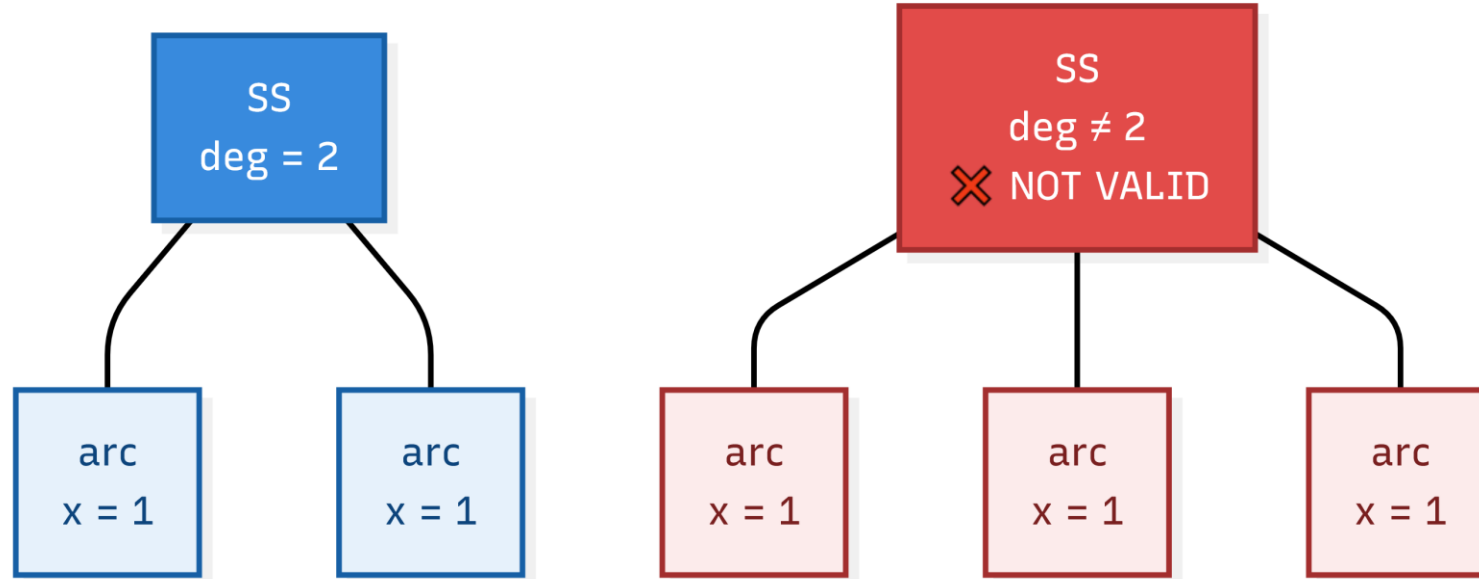
$$-(x_{ij} p_{ij})/2 \leq f_{ij} \leq (x_{ij} p_{ij})/2 \quad \forall (i,j) \in A \quad (11)$$

$$x_{ij} \in \{0, 1\} \quad \forall (i,j) \in A$$

LOOP-FEEDER SUBTOUR ELIMINATION (LF-SE)

$$\sum_{(i,j) \in \delta(k)} x_{ij} = 2 \quad \forall k \in D \quad (7)$$

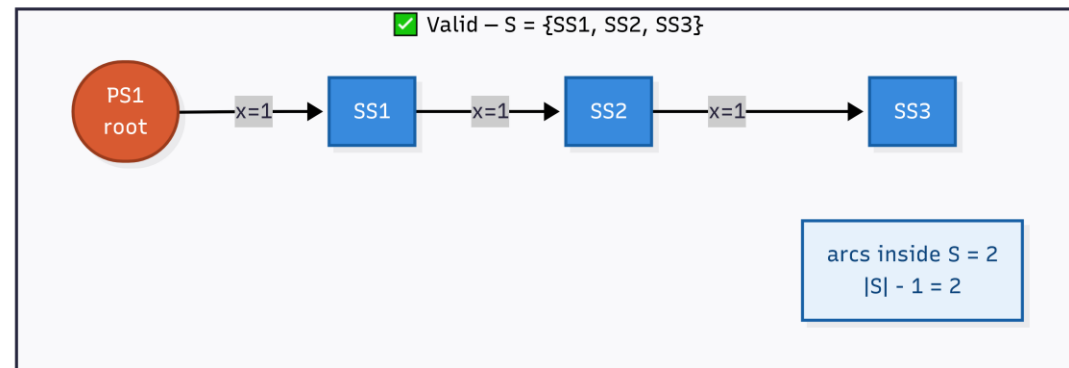
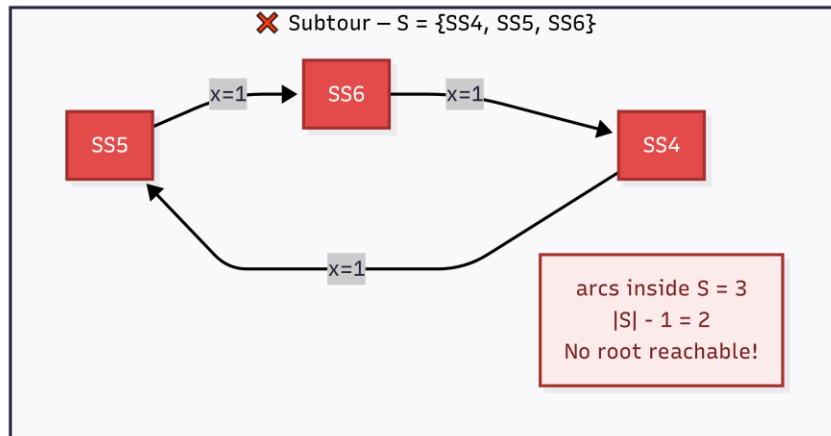
Each demand node has exactly two incident active arcs



LOOP-FEEDER SUBTOUR ELIMINATION (LF-SE)

$$\sum_{(i,j) \in A(S)} x_{ij} \leq |S| - 1 \quad \forall S \subseteq D, |S| \geq 2 \quad (9)$$

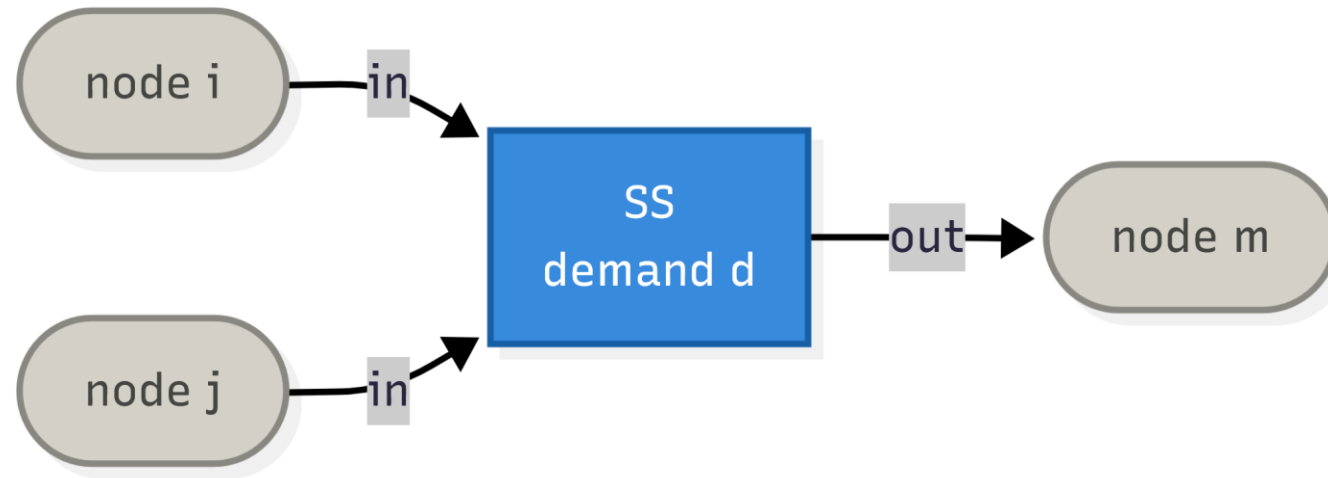
For every subset S of demand nodes with $|S| \geq 2$, the number of active arcs with **both** endpoints inside S must be at most $|S| - 1$.



LOOP-FEEDER SUBTOUR ELIMINATION (LF-SE)

$$\sum_{(i,j) \in \delta^-(k)} f_{ij} = d_k + \sum_{(i,j) \in \delta^+(k)} f_{ij} \quad \forall k \in D \quad (10)$$

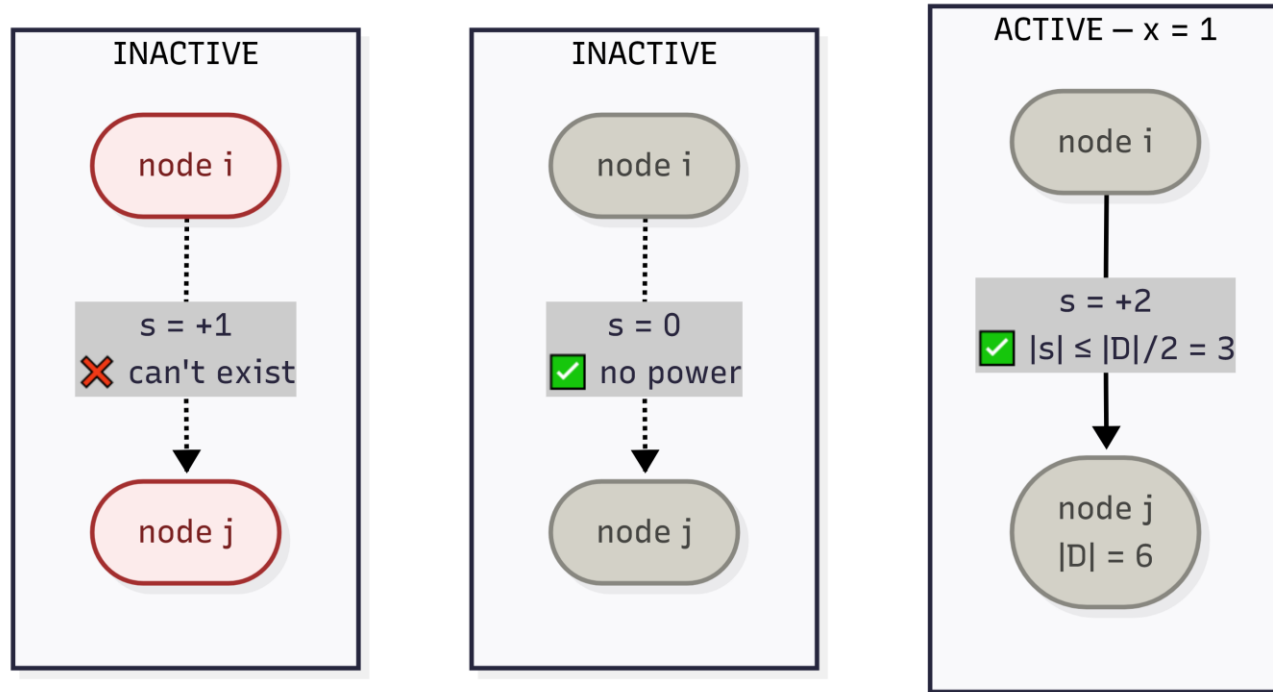
Real power flow conservation: power entering node k equals its demand plus power exiting.



LOOP-FEEDER SUBTOUR ELIMINATION (LF-SE)

$$-(x_{ij}p_{ij})/2 \leq f_{ij} \leq (x_{ij}p_{ij})/2 \quad \forall (i, j) \in A \quad (11)$$

Real power flow is bounded to $\pm$ half capacity if the arc is active, zero otherwise.



SEPARATION PROCEDURE

LF-SE has an **exponentially large** number of SECs, this means it is impossible to enumerate upfront for any realistic instance.

Solution: a **separation-optimisation strategy** that start with no SECs, then dynamically detect and add only the violated ones.

Implemented as a **branch-and-cut routine** with lazy constraint callbacks.

SEPARATION PROCEDURE

A component with no root is an **isolated demand-node cycle**, exactly what the subtour elimination constraint forbids.

1. Find **connected components** of H : For each component S , check whether it contains any root.
2. If S contains **no root**, it is a disconnected subtour and a direct violation.
3. Add the violated SEC as a **lazy constraint** and continue solving.

```
1: Initialization: take the values  $\bar{x}$  from the incumbent solution
   and define the corresponding graph  $H(N, A(\bar{x}))$ 
2: Find connected components:
   connComponents  $\leftarrow$  PartitionIntoConnComp( $H$ )
3: for each component in connComponents do
4:    $S \leftarrow$  component  $\triangleright$  get the current component
5:   if  $|S \cap R| = 0$  then
6:     Compute newConstraint
7:     AddLazyConstraint(newConstraint)
8:   end if
9: end for
```

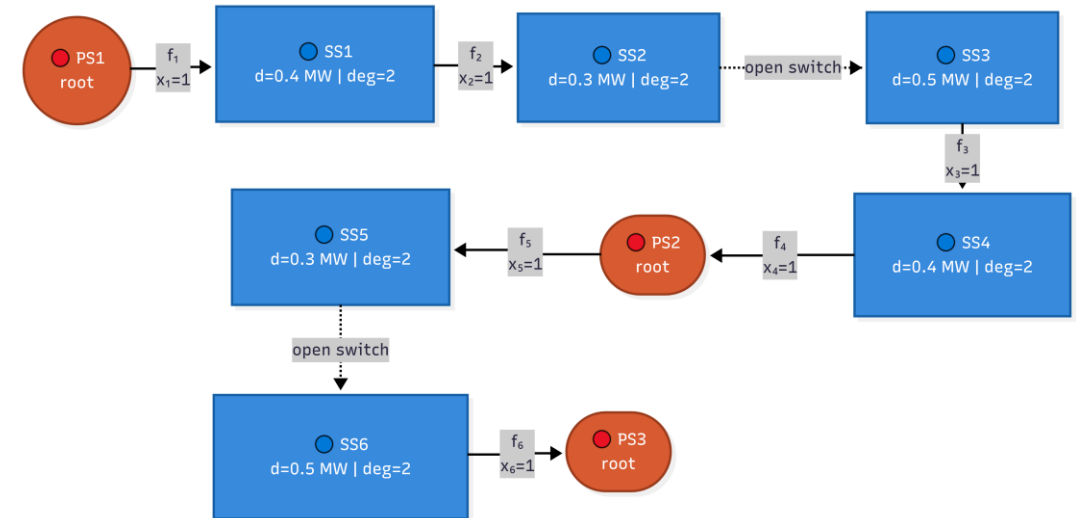

Loop-Feeder Formulations: Comparison

	LF-SCF	LF-SE
Connectivity enforcement	Fictitious flow variables s	Subtour Elimination Constraints (9)
Extra variables	$s(i, j)$ for every arc	None
Number of constraints	Polynomial	Exponential
Topology produced	Loop-feeder	Loop-feeder

OPEN-LOOP SUBTOUR ELIMINATION (OL-SE)

Extends LF-SE to enforce the stricter **open-loop topology**, where every power line must connect **two different PSs**.

Same objective and power flow constraints as LF-SE; differs only in the **scope of subtour elimination** because is extended to subsets containing at most one root.



OPEN-LOOP SUBTOUR ELIMINATION (OL-SE)

$$\min \sum_{(i,j) \in A} c_{ij} x_{ij} \quad (12)$$

s.t.

$$\sum_{(i,j) \in \delta(k)} x_{ij} = 2 \quad \forall k \in D \quad (13)$$

$$\sum_{(i,j) \in A(S)} x_{ij} \leq |S| - 1 \quad \forall S \subseteq N, |S| \geq 2, |S \cap R| \leq 1 \quad (14)$$

$$\sum_{(i,j) \in \delta^-(k)} f_{ij} = d_k + \sum_{(i,j) \in \delta^+(k)} f_{ij} \quad \forall k \in D \quad (15)$$

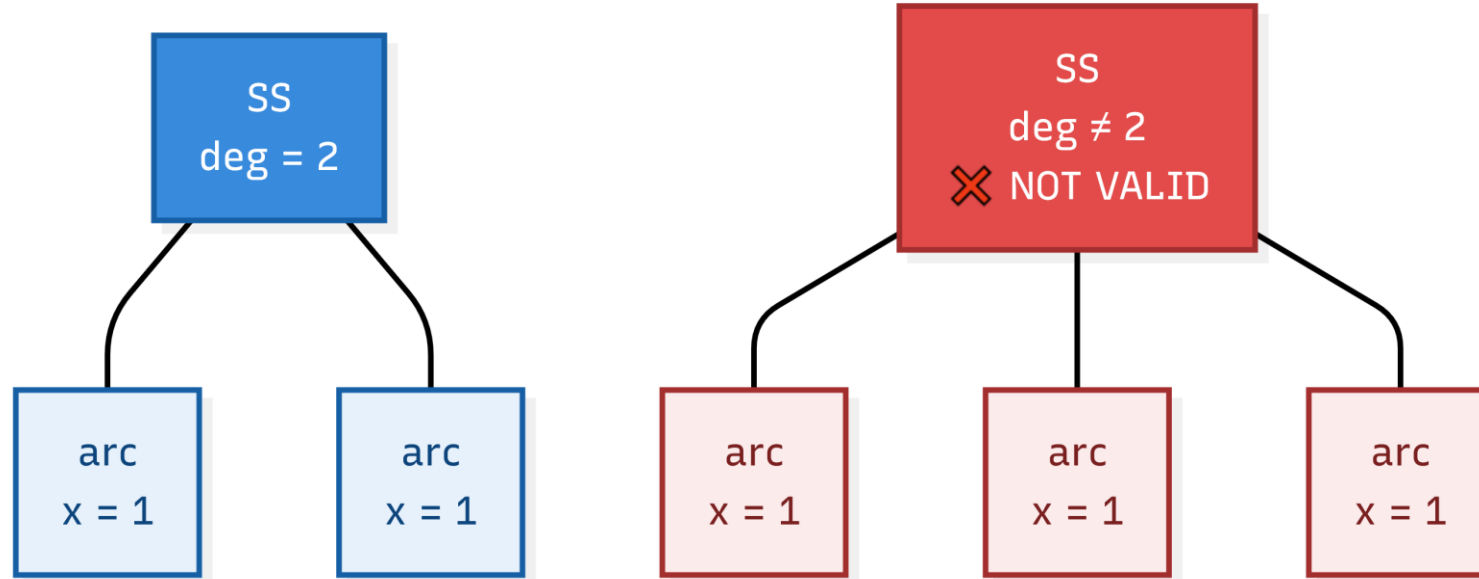
$$-(x_{ij} p_{ij})/2 \leq f_{ij} \leq (x_{ij} p_{ij})/2 \quad \forall (i,j) \in A \quad (16)$$

$$x_{ij} \in \{0, 1\} \quad \forall (i,j) \in A$$

OPEN-LOOP SUBTOUR ELIMINATION (OL-SE)

$$\sum_{(i,j) \in \delta(k)} x_{ij} = 2 \quad \forall k \in D \quad (13)$$

Each demand node has exactly two incident active arcs

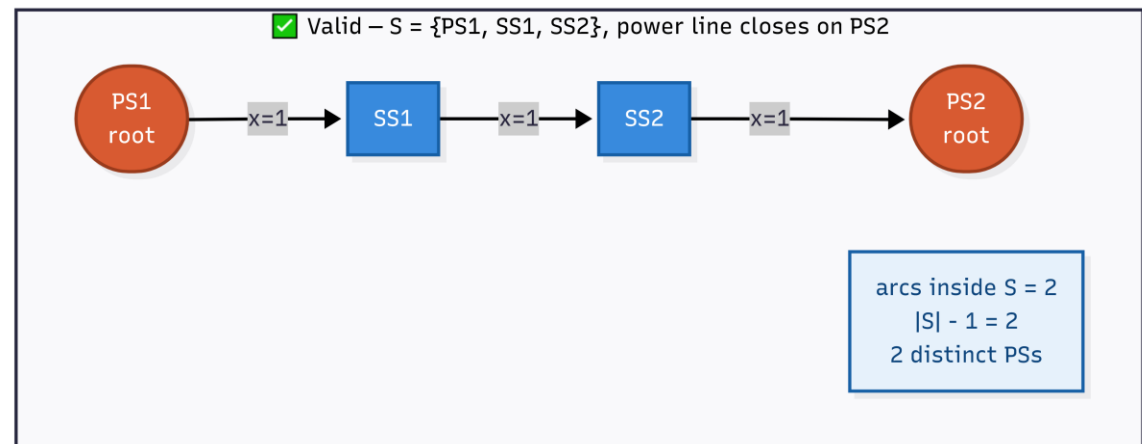
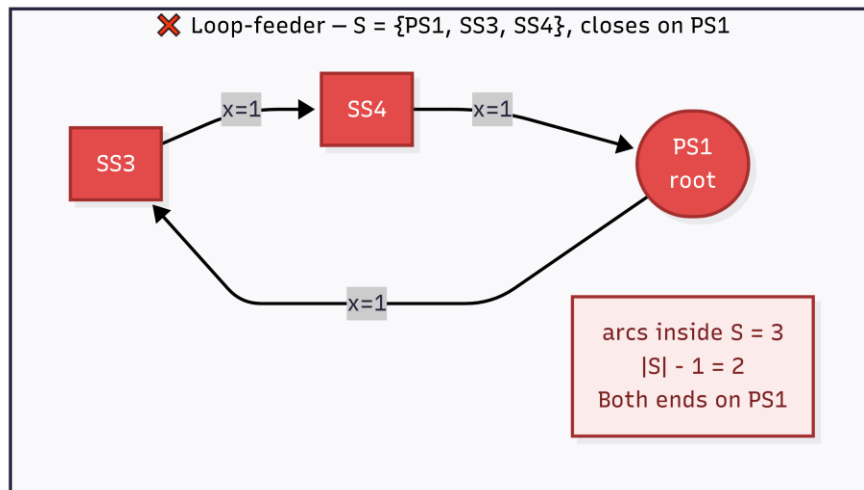


OPEN-LOOP SUBTOUR ELIMINATION (OL-SE)

$$\sum_{(i,j) \in A(S)} x_{ij} \leq |S| - 1 \quad \forall S \subseteq N, |S| \geq 2, |S \cap R| \leq 1 \quad (14)$$

Extended Subtour Elimination from the LF-SE. This extends the elimination to **any subset S containing at most one root $|S \cap R| \leq 1$** .

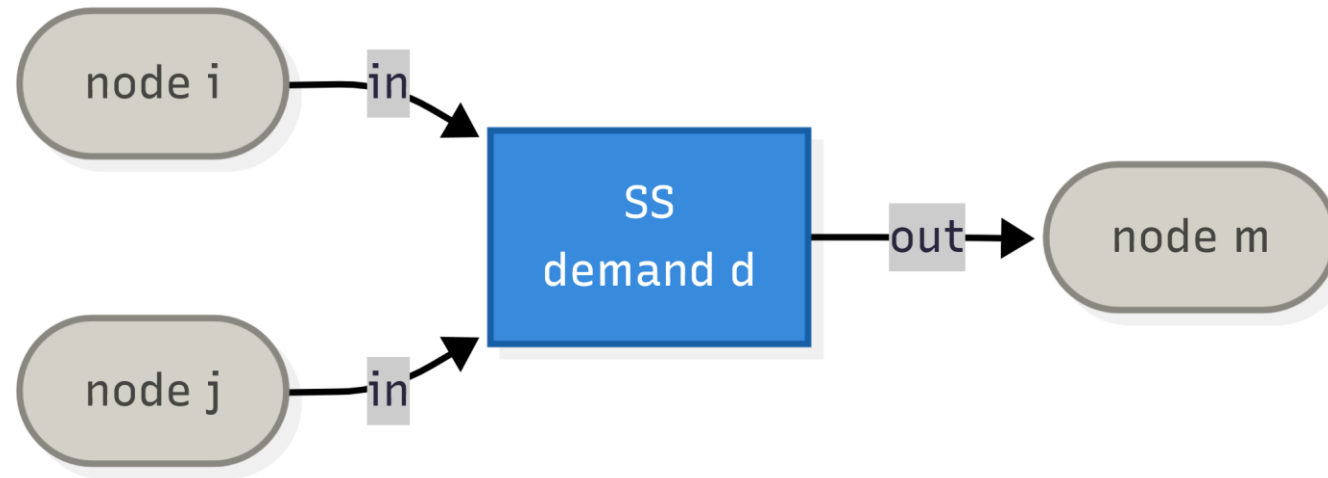
This forces every SS to be reachable from **two distinct PS s**, ruling out any power line that loops back to the same root.



OPEN-LOOP SUBTOUR ELIMINATION (OL-SE)

$$\sum_{(i,j) \in \delta^-(k)} f_{ij} = d_k + \sum_{(i,j) \in \delta^+(k)} f_{ij} \quad \forall k \in D \quad (15)$$

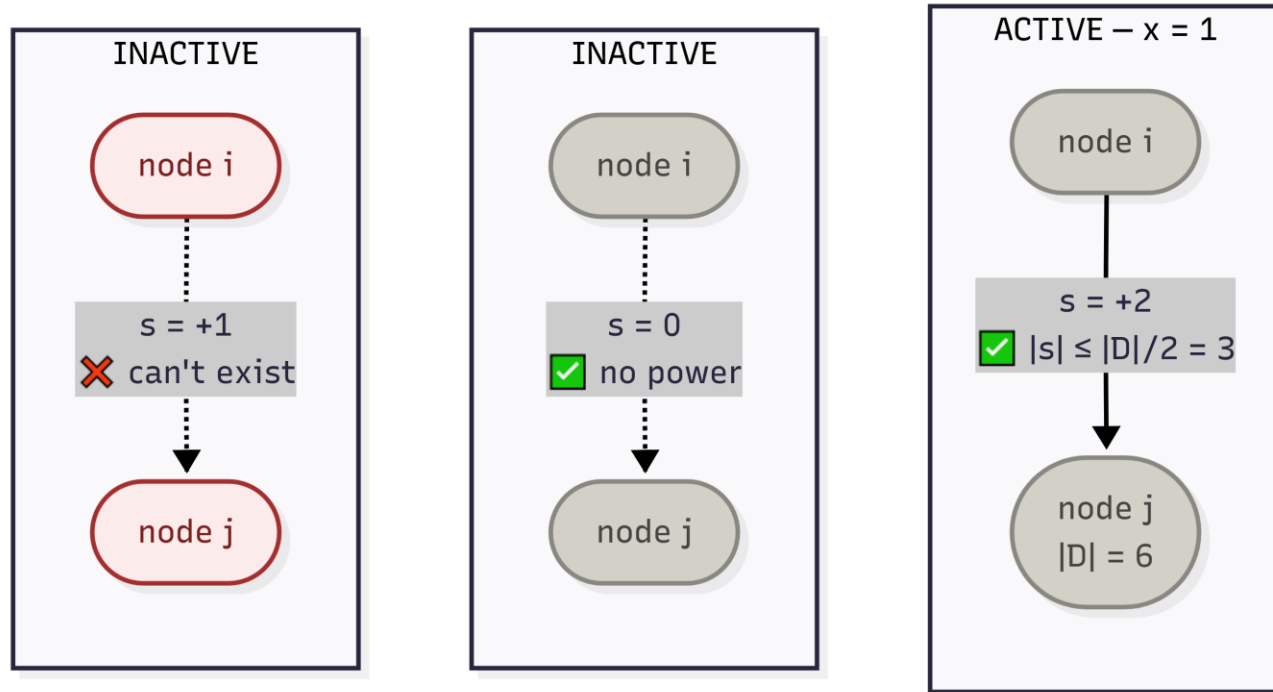
Real power flow conservation: power entering node k equals its demand plus power exiting.



OPEN-LOOP SUBTOUR ELIMINATION (OL-SE)

$$-(x_{ij}p_{ij})/2 \leq f_{ij} \leq (x_{ij}p_{ij})/2 \quad \forall (i, j) \in A \quad (16)$$

Real power flow is bounded to $\pm$ half capacity if the arc is active, zero otherwise.



SEPARATION PROCEDURE

Constraint (14) applies to subsets containing **at most one root**. This is not a pure demand subtours. An **augmented graph H^*** is built by adding a virtual node n^* adjacent to all roots.

Steps:

- Build H^* and add n^* adjacent to every root.
- Find **biconnected components** via Tarjan's algorithm.
- If a component contains ≤ 1 root, it is a violation.
- Add the violated SEC as a **lazy constraint** and continue.

1: Initialization: take the values $\bar{x}$ from the incumbent solution and define the corresponding graph $H(N^*, A^*(\bar{x}))$

2: Find biconnected components:

$biConnComponents \leftarrow PartitionIntoBiConnComp(H)$

3: for each *bicomponent* in *biConnComponents* do

4: $S \leftarrow bicomponent \triangleright$ get the current biconnected component

5: if $|S \cap R| \leq 1$ and $n^* \notin S$ then

6: *Compute newConstraint*

7: *AddLazyConstraint(newConstraint)*

8: end if

9: end for

	LF-SCF	LF-SE	OL-SE
Connectivity enforcement	Fictitious flow (4)	SECs on D only (9)	SECs on N, ≤ 1 root (14)
Extra variables	$s(i, j)$	None	None
Constraint count	Polynomial	Exponential	Exponential
Topology	Loop-feeder	Loop-feeder	Open-loop
PS fault resilient	NO	NO	YES

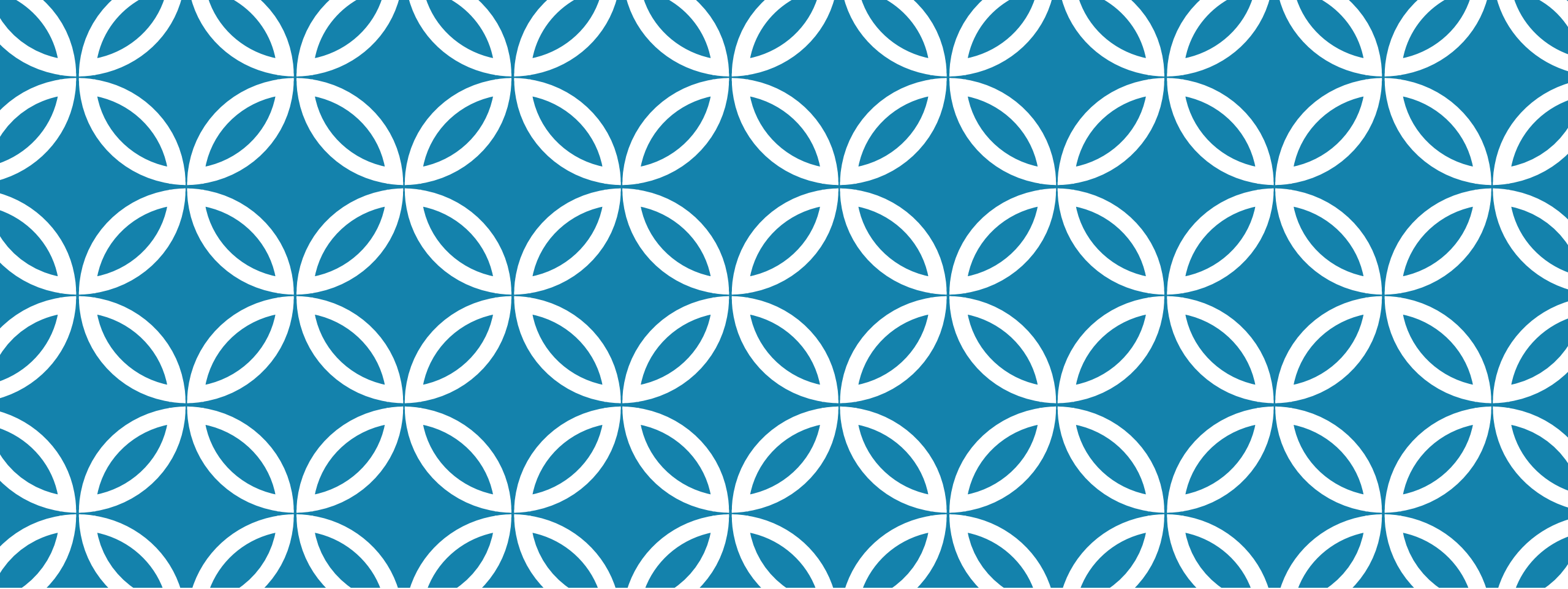
CUT CONSTRAINTS

The research also compares the results with a cut-based constraints on top of the existing OL-SE formulation.

OL-SE+CC adds an additional family of explicit cut constraints upfront at the start of the optimization, for every combination of demand node subset S and root $r \in R$:

$$\sum_{e \in \delta(S', T)} x_e \geq \sum_{e \in \delta(S', r)} x_e \quad \forall S' \subseteq D, r \in R$$

The number of arcs leaving S' toward the rest of the network must be at least as large as the number connecting S' back to root r , preventing any power line from looping on a single PS.



EXPERIMENTAL RESULTS

DATA SOURCE

There are no external benchmark dataset used. The project needs to statically generate the NetworkData.

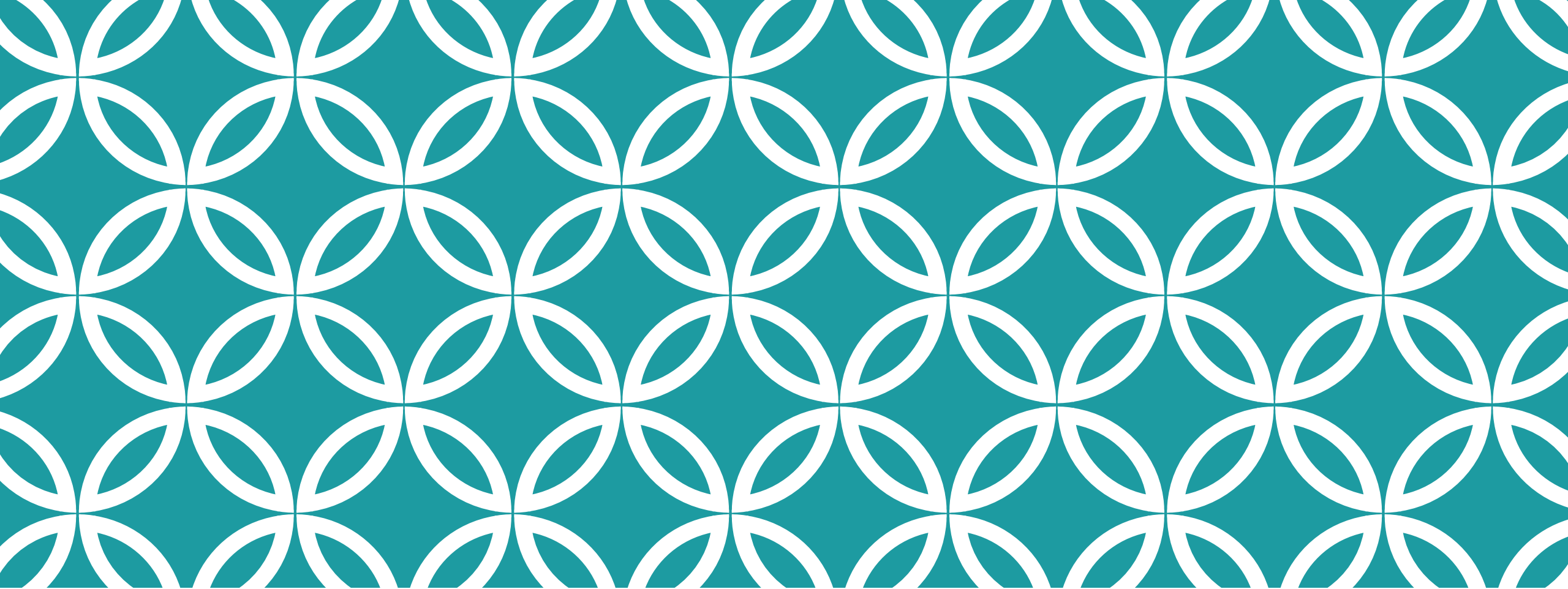
The code inside the generator.py statically builds the network from given parameters and a SEED.

Case study	PSs		SSs		Density (nodes/km2)	
	Center	Suburbs	Center	Suburbs	Center	Suburbs
Case 54	2	2	25	25	3,00	0,74
Case 78	1	2	30	45	7,75	3,92
Case 104	1	3	60	40	7,63	1,59
Case 154	2	2	80	70	6,83	1,50

Cost per km	$30.000 \cdot (1 + c)$ €/km
Cost obstacle	1.000.000€
Max Extra Vio	0,1
Center mean	0,40MW
Center dev.	0,20MW
Suburb mean	0,20MW
Suburb dev.	0,60MW

HARDWARE SPECIFICATIONS

CPU	Intel i9-10850K (10 core, 20 thread) @ 4.500GHz
RAM	64GB
ENV	Python 3.12.3, Gurobi 13
System	64 bit on x64

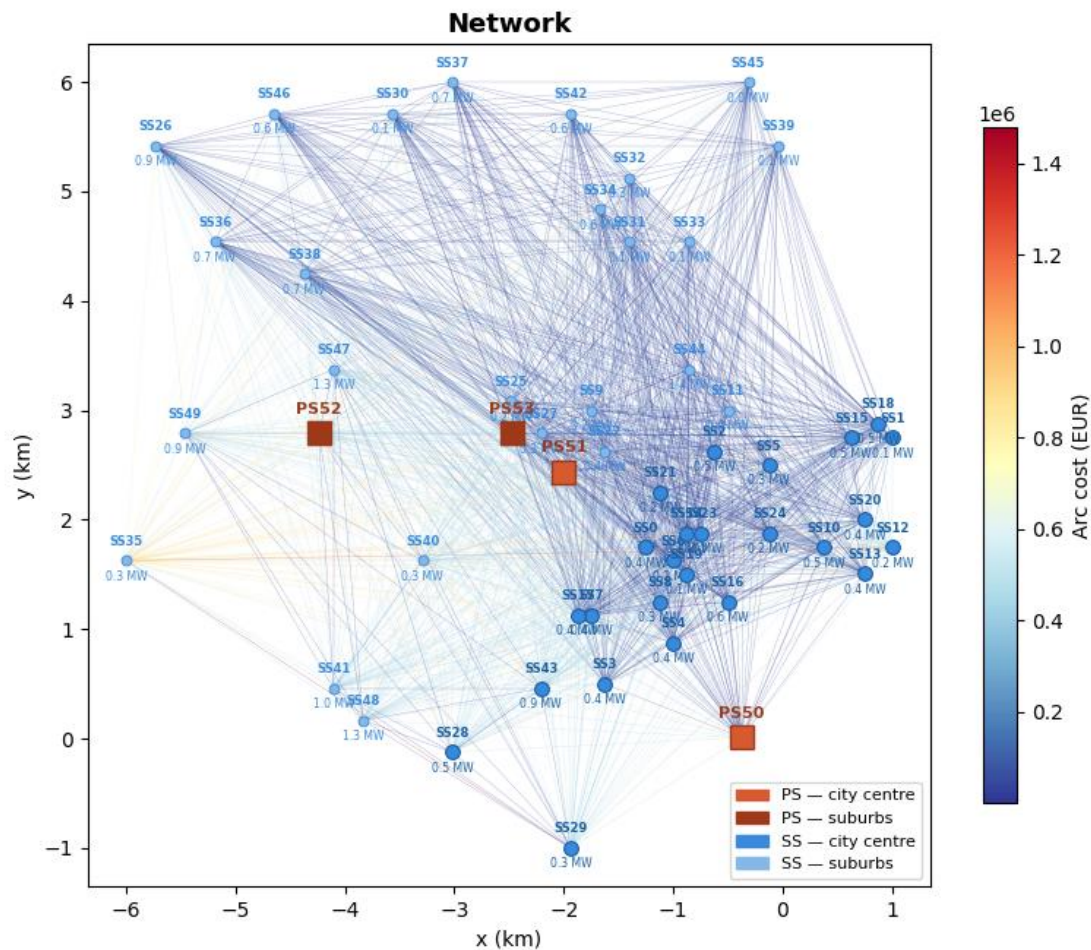
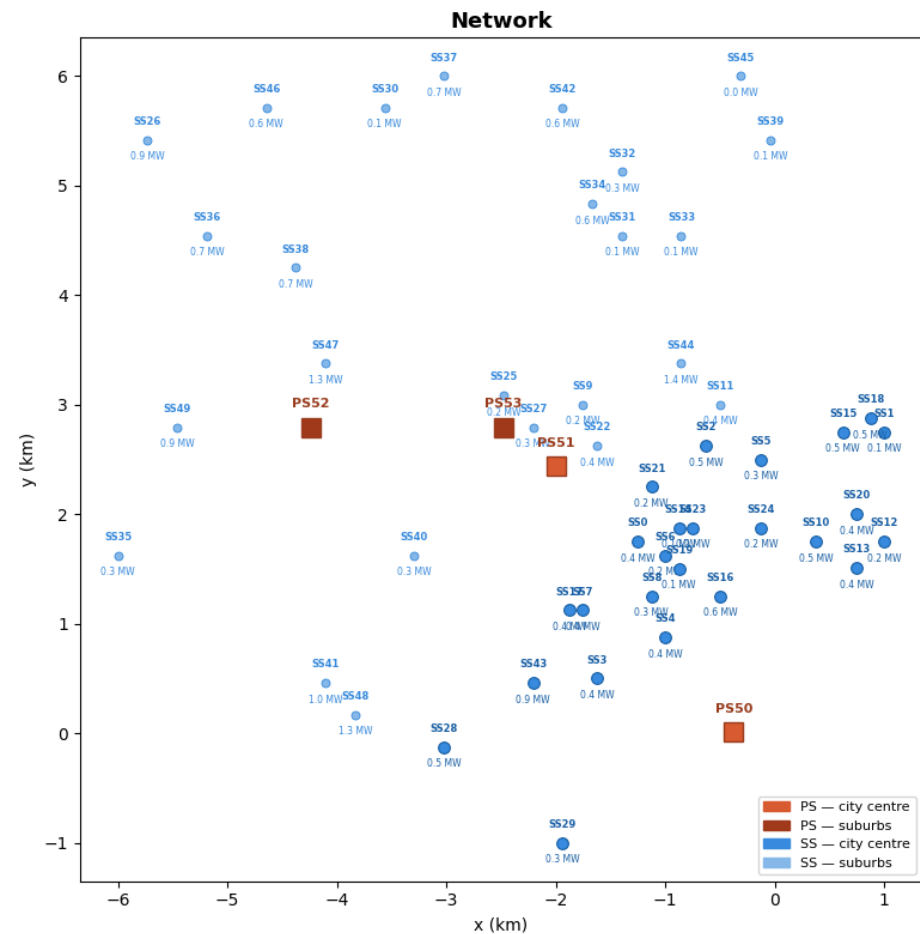


DATASET

The paper provides some pre-generated network datasets

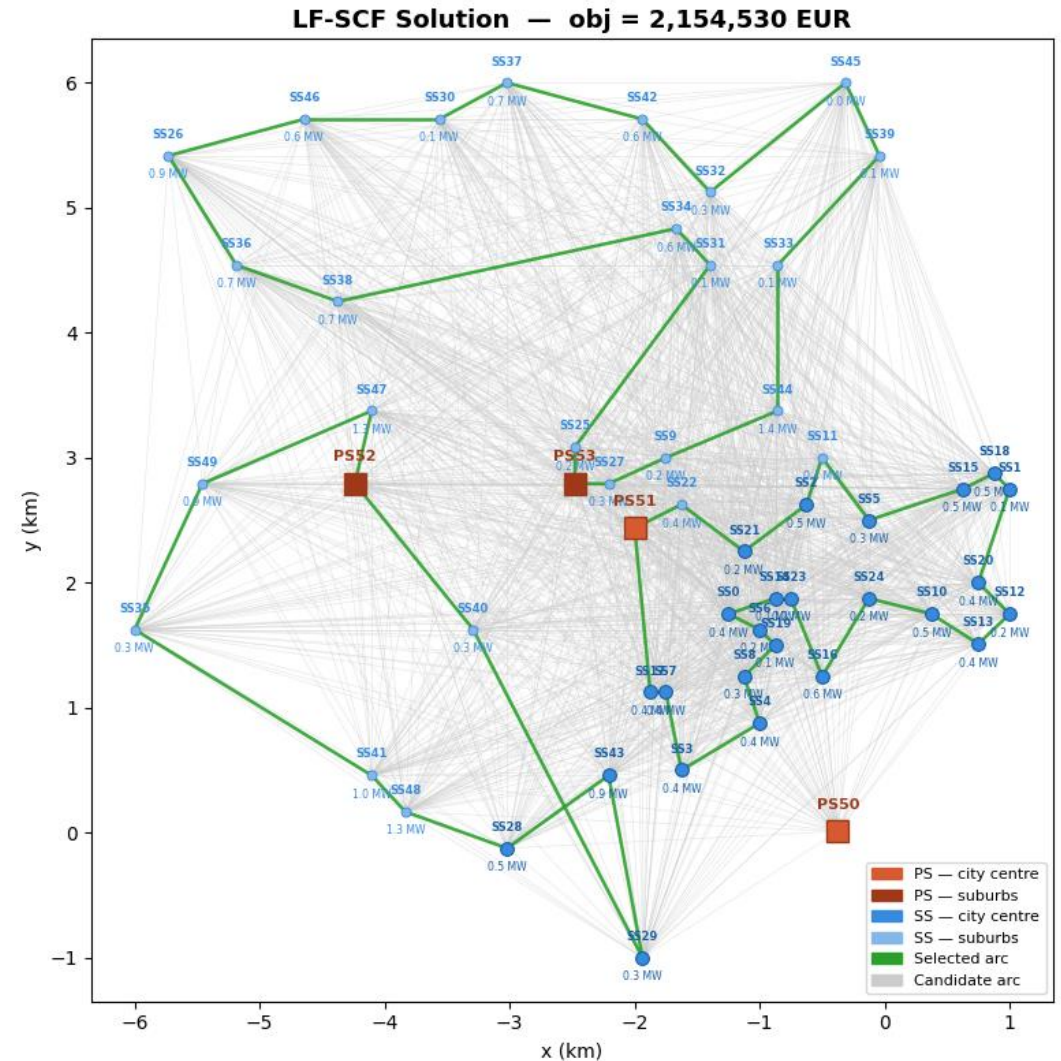
DATA SOURCE

Case study	PSs		SSs		Density (nodes/km2)	
	Center	Suburbs	Center	Suburbs	Center	Suburbs
Case 54	2	2	25	25	3,00	0,74



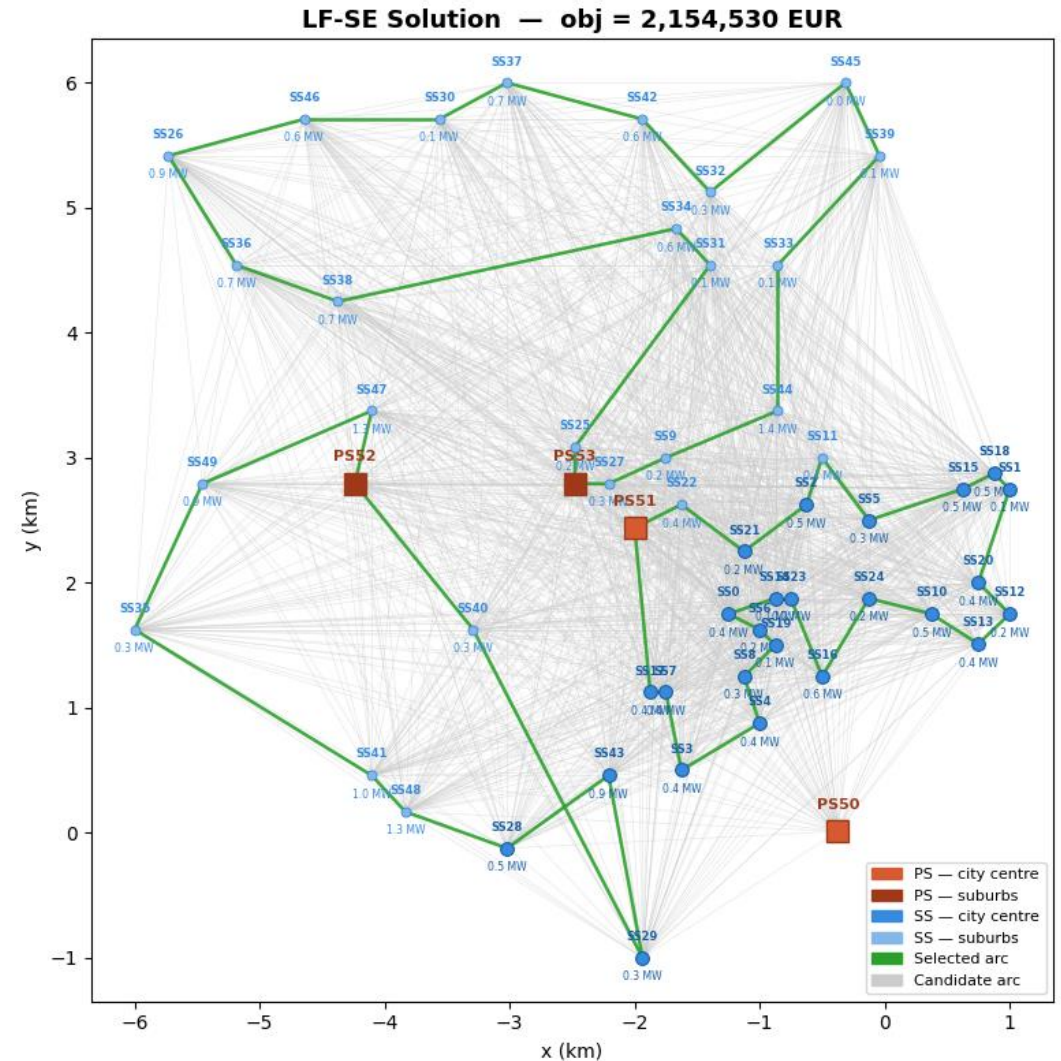
SOLUTION LF-SCF

#NODES	54
#ARCS	1431
Cable capacity	10 MW
Total demand	22.44 MW
Execution time	2.65s
Gap	0.00%
Root gap	2.91%
Max execution time	60m
Solution	2,154,530



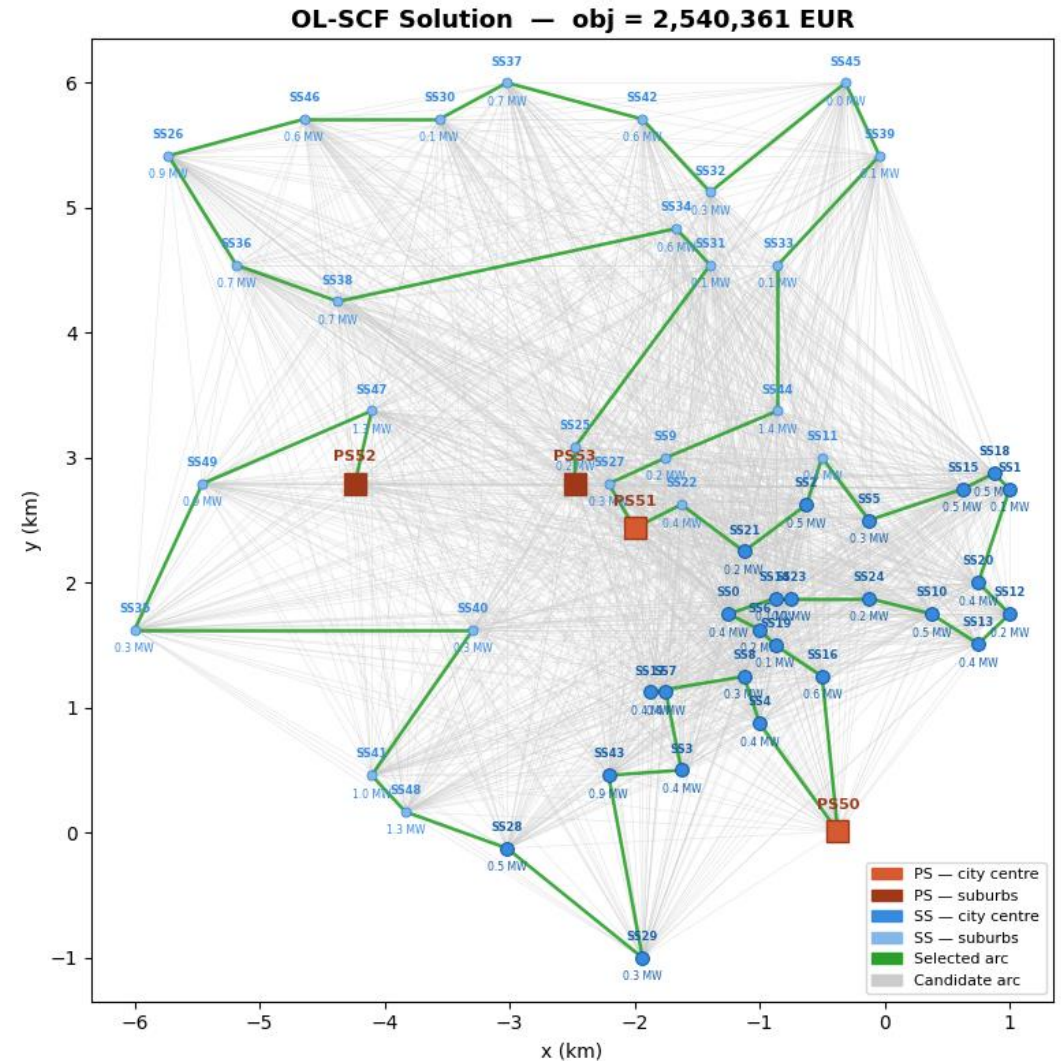
SOLUTION LF-SE

#NODES	54
#ARCS	1431
Cable capacity	10 MW
Total demand	22.44 MW
Execution time	1.10s
Gap	0.00%
Root gap	2.95%
Max execution time	60m
Solution	2,154,530



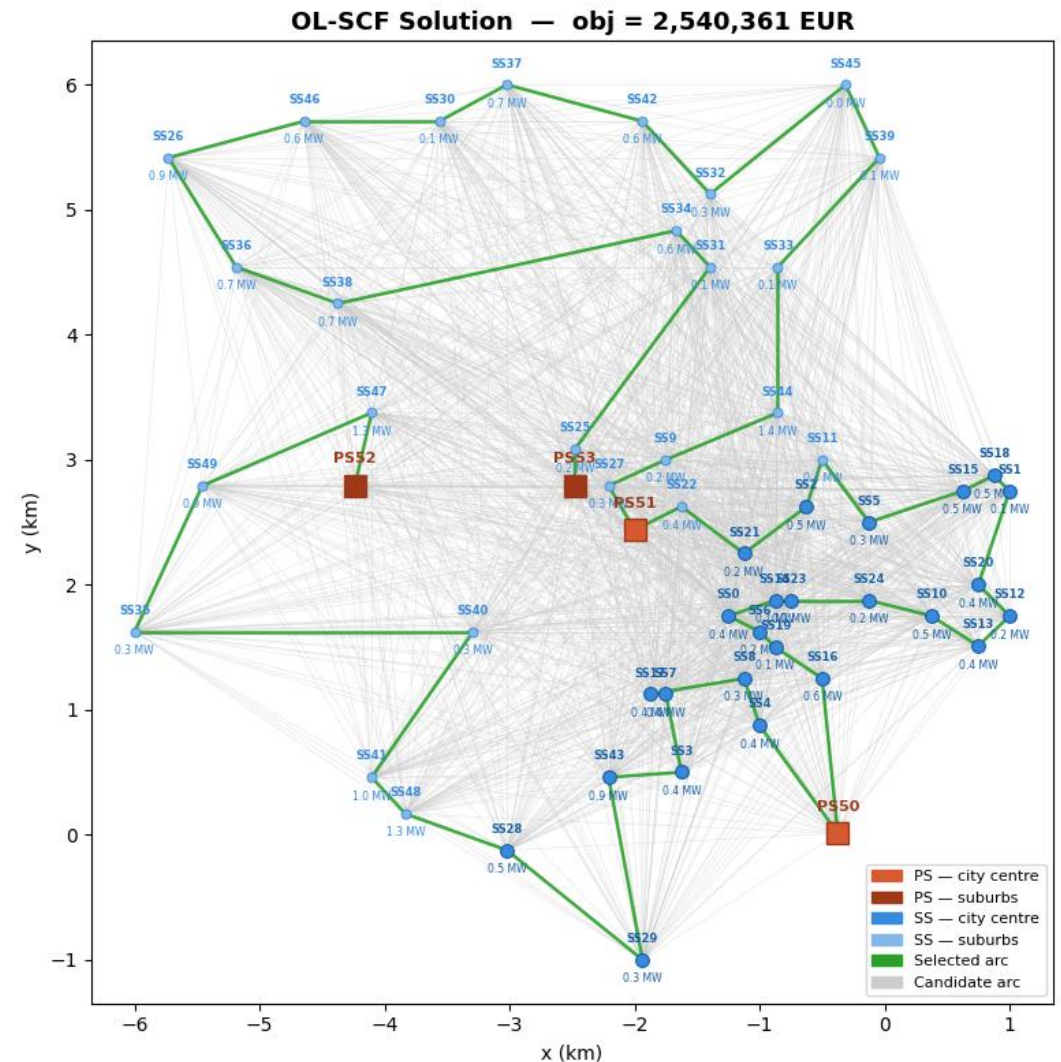
SOLUTION OL-SE

#NODES	54
#ARCS	1431
Cable capacity	10 MW
Total demand	22.44 MW
Execution time	22.42s
Gap	0.00%
Root gap	17.69%
Max execution time	60m
Solution	2,540,361



SOLUTION OL-SE+CC

#NODES	54
#ARCS	1431
Cable capacity	10 MW
Total demand	22.44 MW
Execution time	19.08s
Gap	0.00%
Root gap	17.69%
Max execution time	60m
Solution	2,540,361



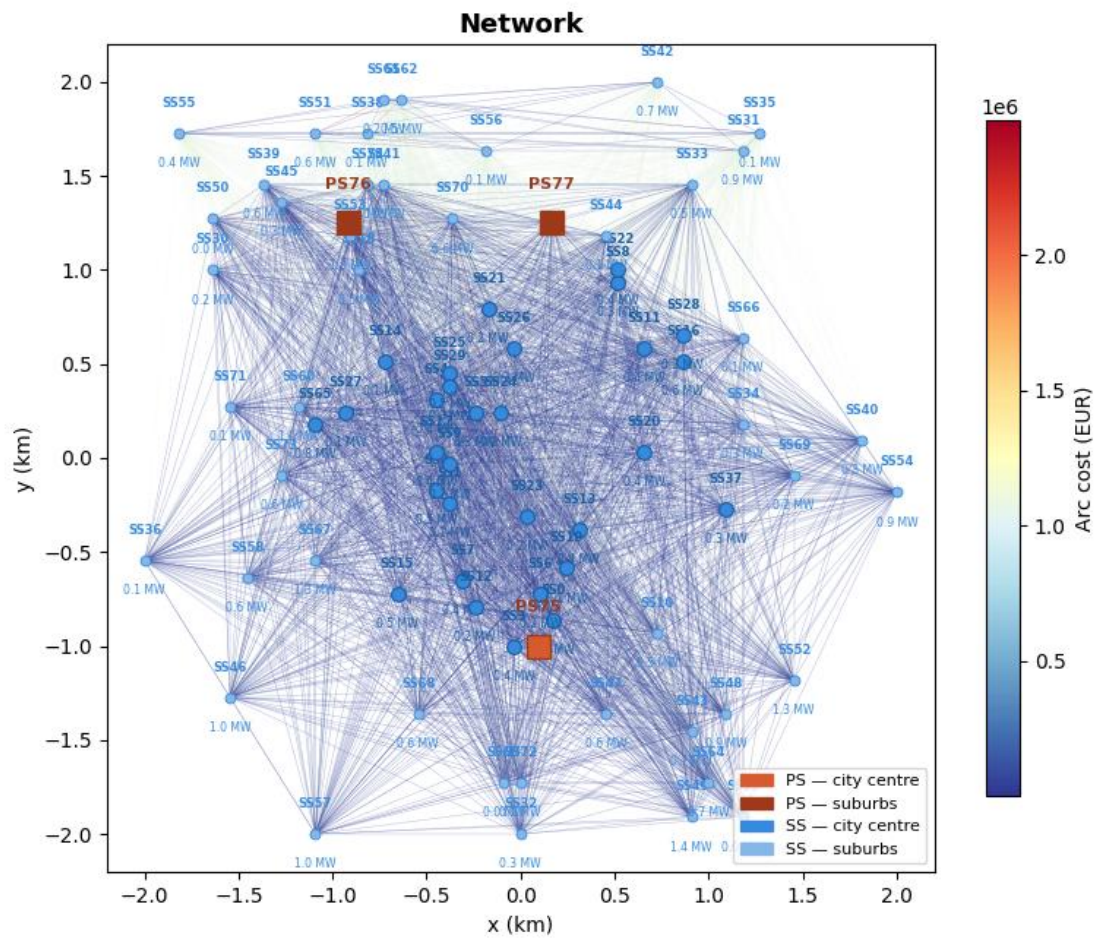
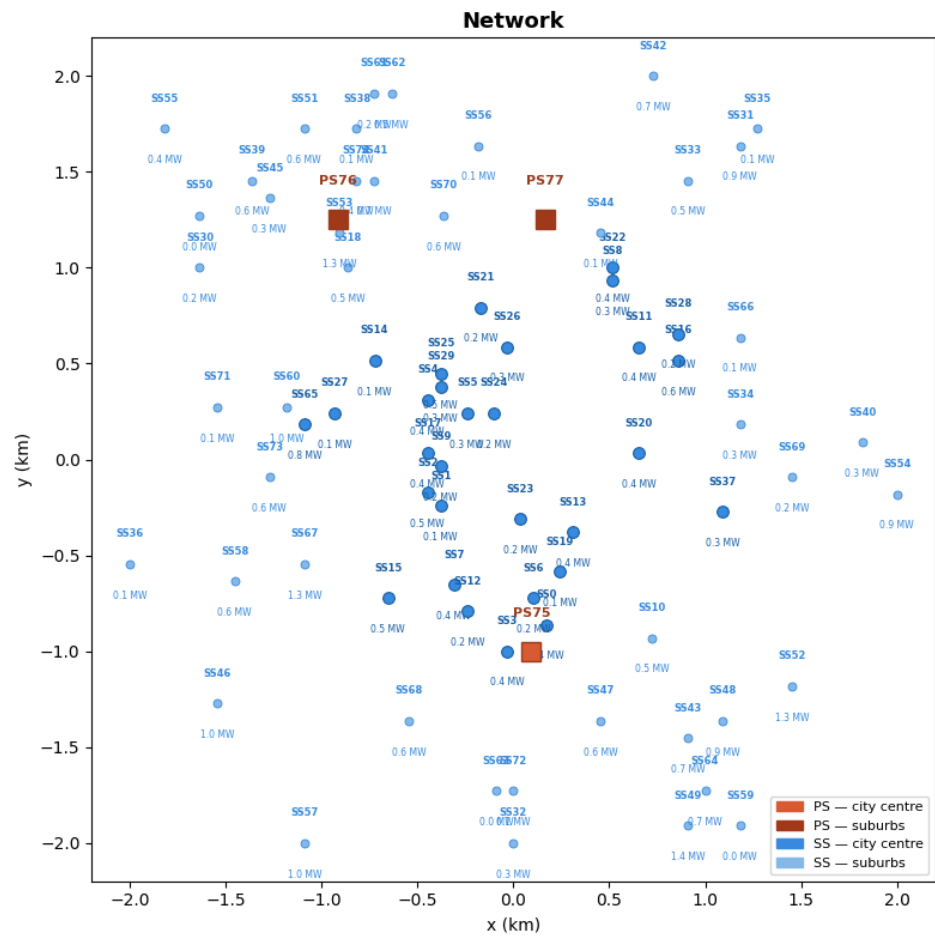
SOLUTION CASE54

#NODES	54
#ARCS	1431
Cable capacity	10 MW
Total demand	22.44 MW

Problem	Execution time	Gap	Root gap	Max execution time	Solution
LF-SCF	2.65s	0.00%	2.91%	60m	2,154,530
LF-SE	1.10s	0.00%	2.95%	60m	2,154,530
OL-SE	22.42s	0.00%	17.69%	60m	2,540,361
OL-SE+CC	19.08s	0.00%	17.69%	60m	2,540,361

DATA SOURCE

Case study	PSs		SSs		Density (nodes/km2)	
	Center	Suburbs	Center	Suburbs	Center	Suburbs
Case 78	1	2	30	45	7,75	3,92

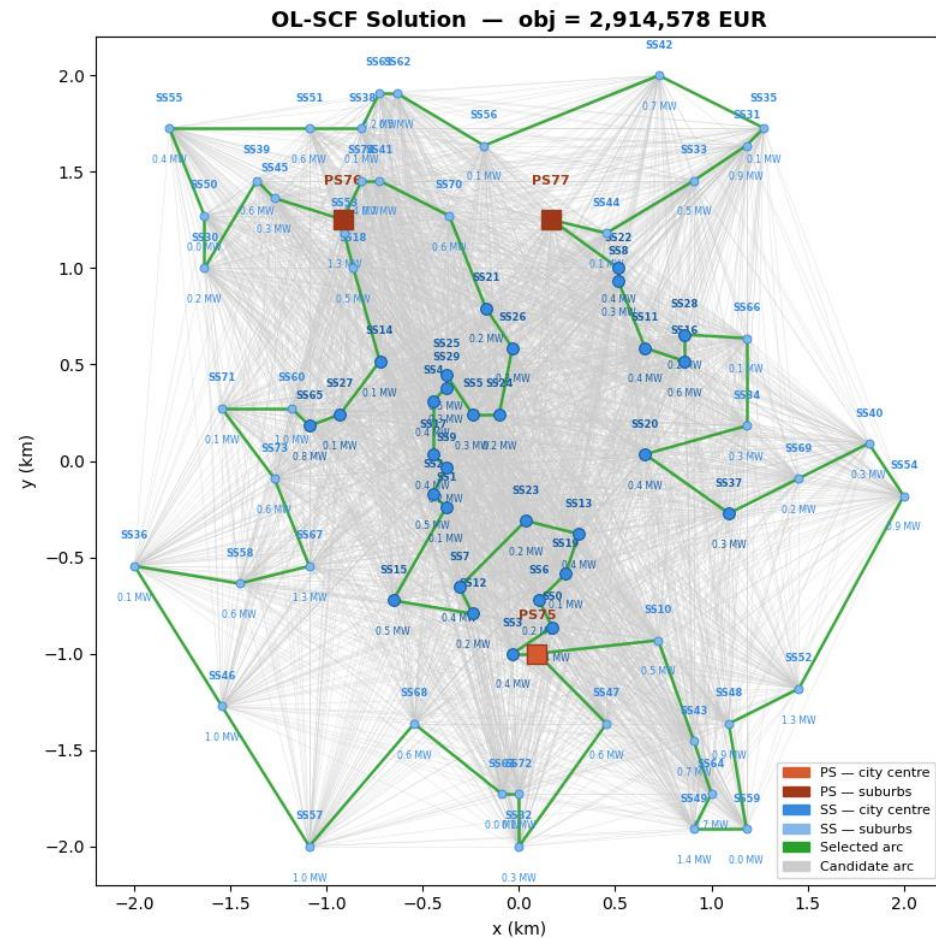
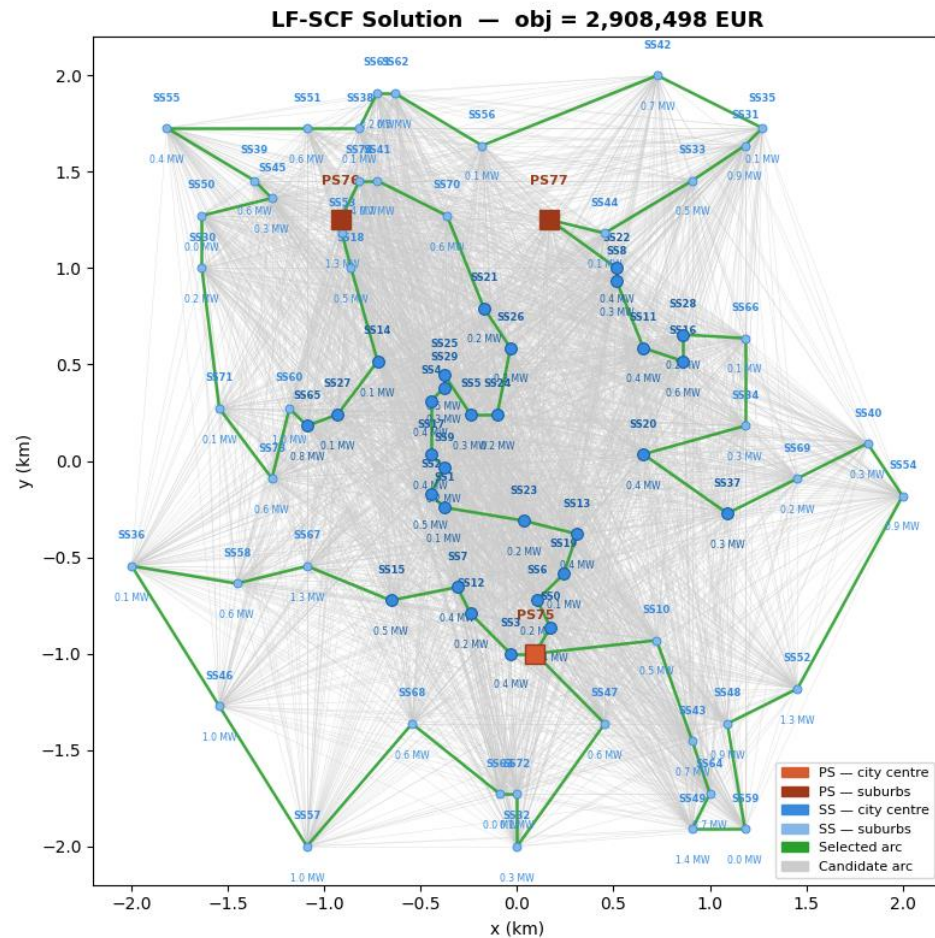


SOLUTION CASE78

#NODES	78
#ARCS	3003
Cable capacity	10 MW
Total demand	32.97 MW

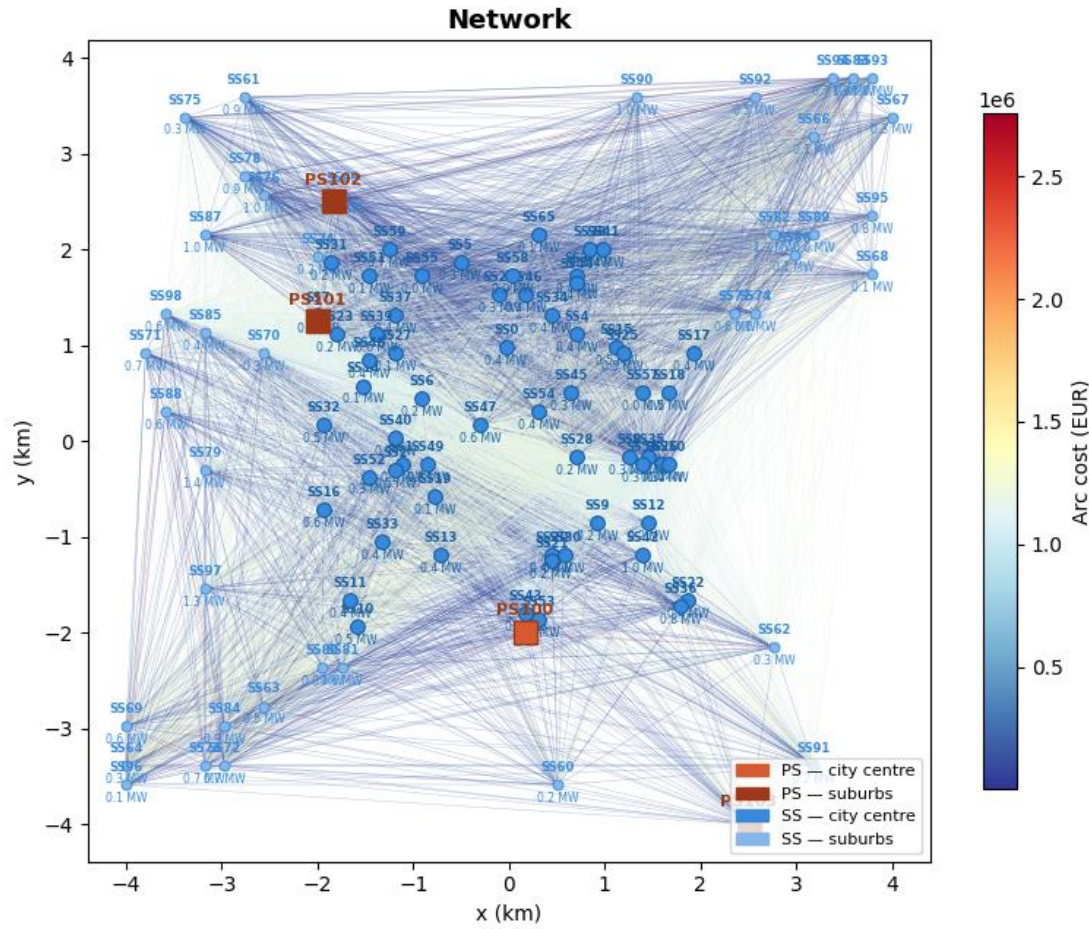
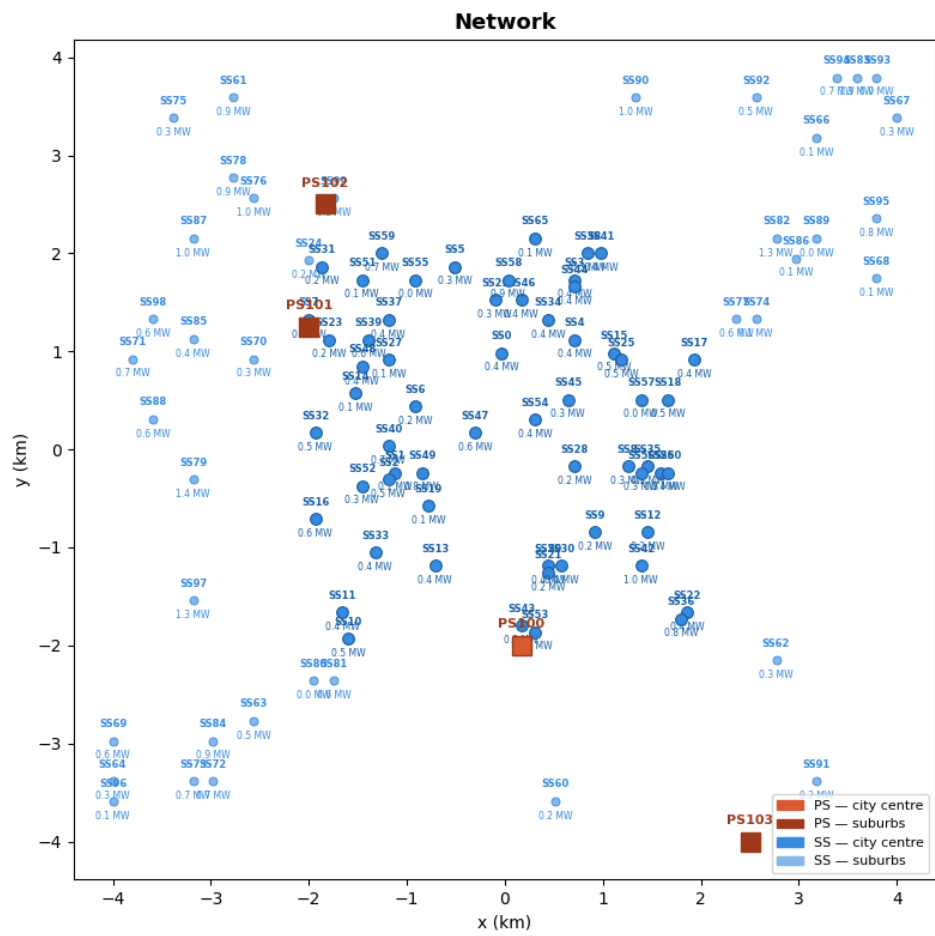
Problem	Execution time	Gap	Root gap	Max execution time	Solution
LF-SCF	21.49s	0.00%	45.71%	60m	2,908,498
LF-SE	17.07s	0.00%	45.71%	60m	2,908,498
OL-SE	55.36s	0.00%	45.72%	60m	2,914,578
OL-SE+CC	90.21s	0.00%	45.72%	60m	2,914,578

SOLUTION CASE78



DATA SOURCE

Case study	PSs		SSs		Density (nodes/km2)	
	Center	Suburbs	Center	Suburbs	Center	Suburbs
Case 104	1	3	60	40	7,63	1,59

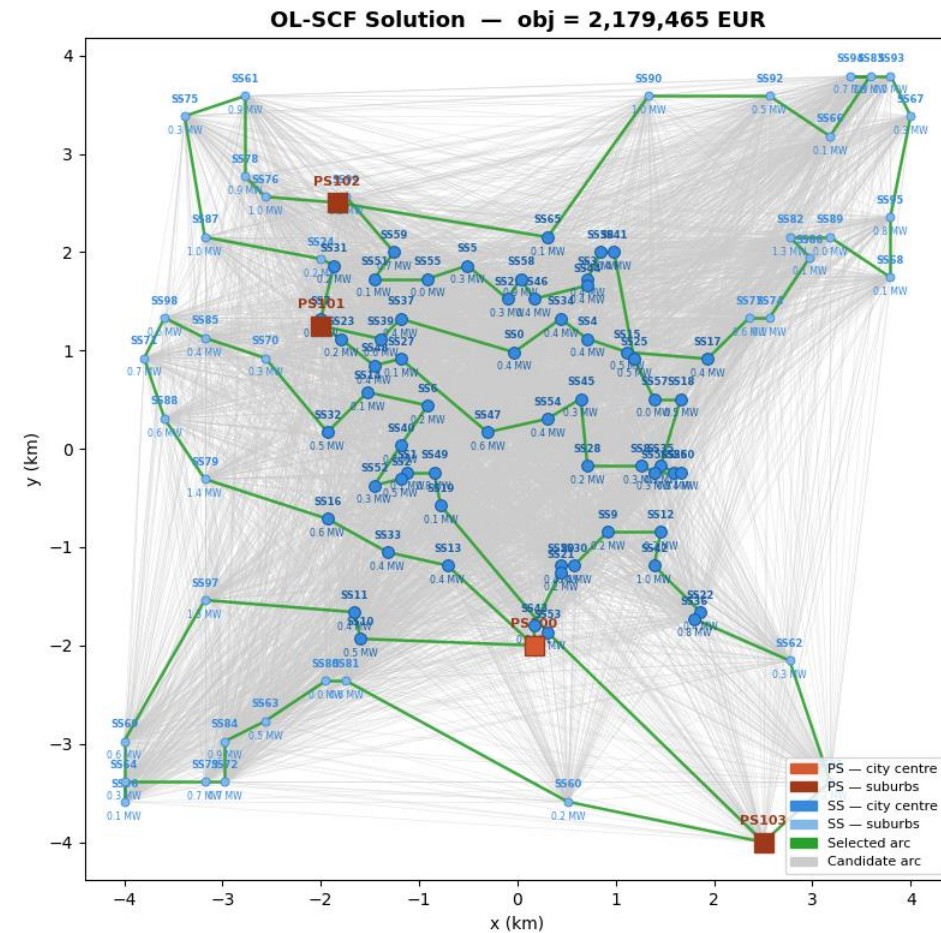
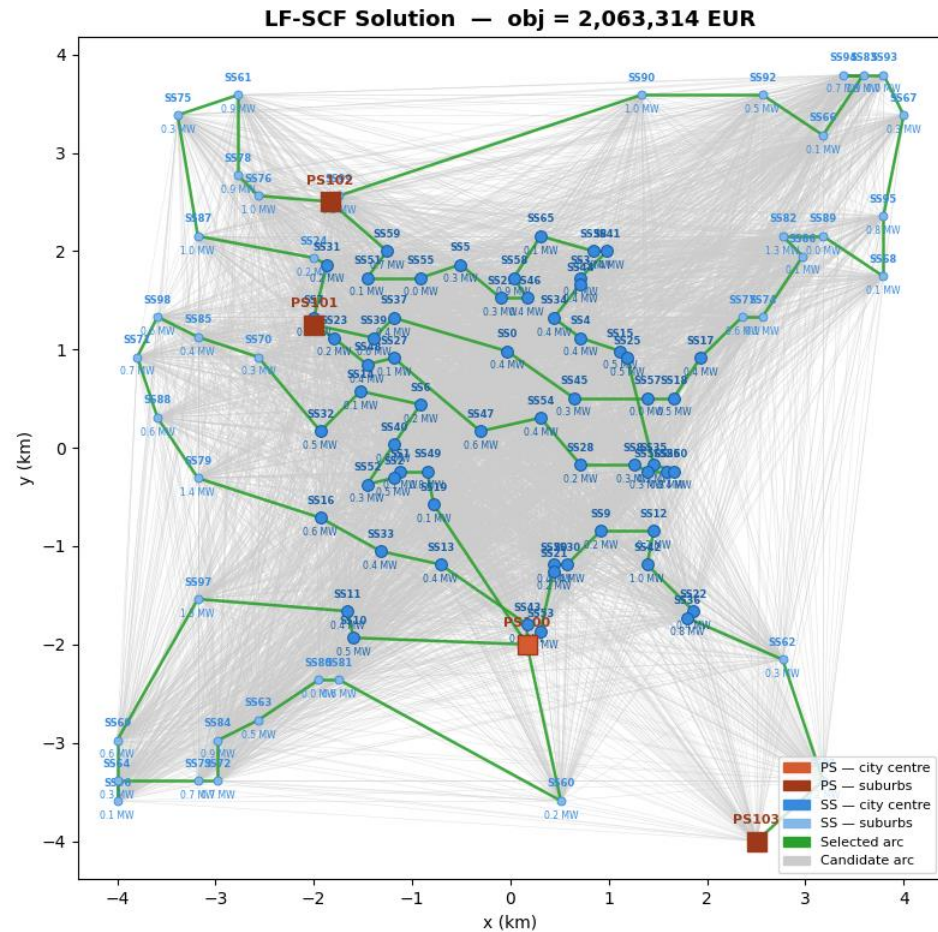


SOLUTION CASE104

#NODES	104
#ARCS	5356
Cable capacity	10 MW
Total demand	32.97 MW

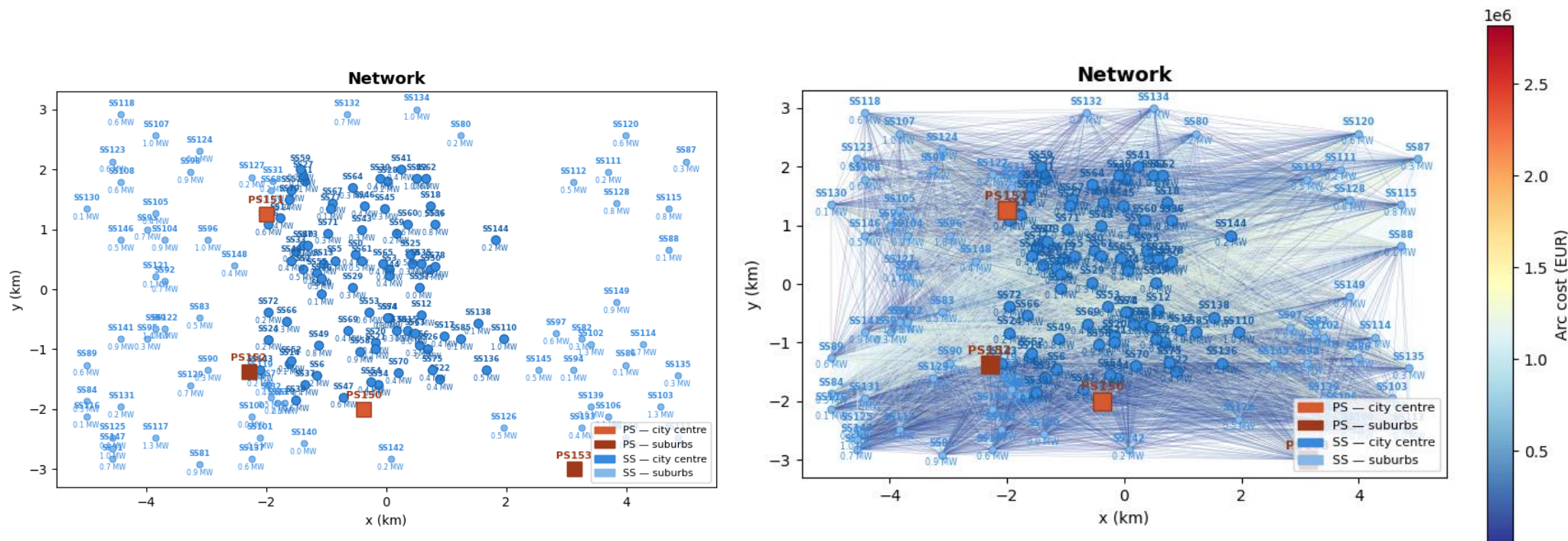
Problem	Execution time	Gap	Root gap	Max execution time	Solution
LF-SCF	Limit reached	0.48%	9.85%	60m	2,063,314
LF-SE	Limit reached	3.00%	10.05%	60m	2,067,886
OL-SE	Limit reached	9.27%	14.56%	60m	2,179,465
OL-SE+CC	Limit reached	9.86%	15.85%	60m	2,210,492

SOLUTION CASE



DATA SOURCE

Case study	PSs		SSs		Density (nodes/km2)	
	Center	Suburbs	Center	Suburbs	Center	Suburbs
Case 154	2	2	80	70	6,83	1,50

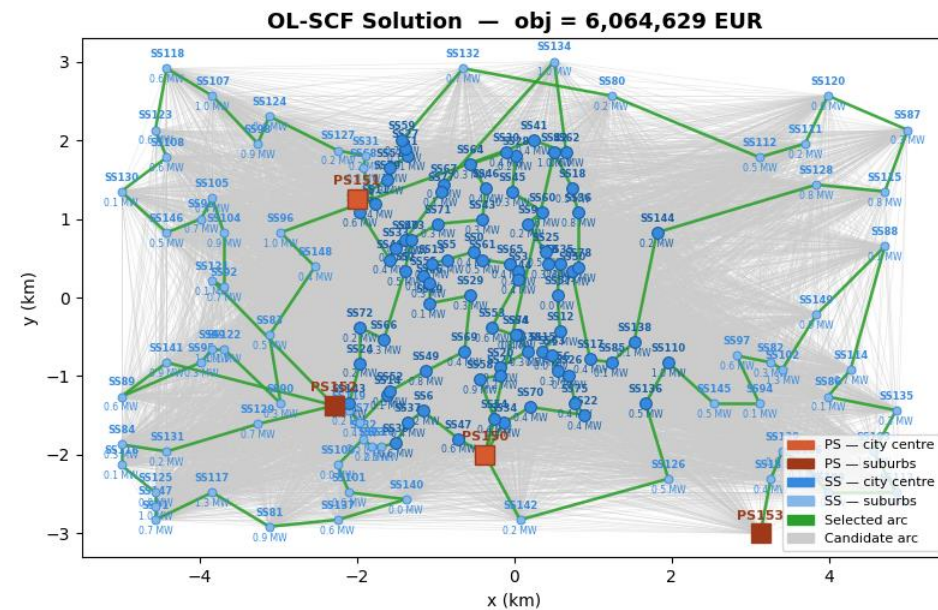
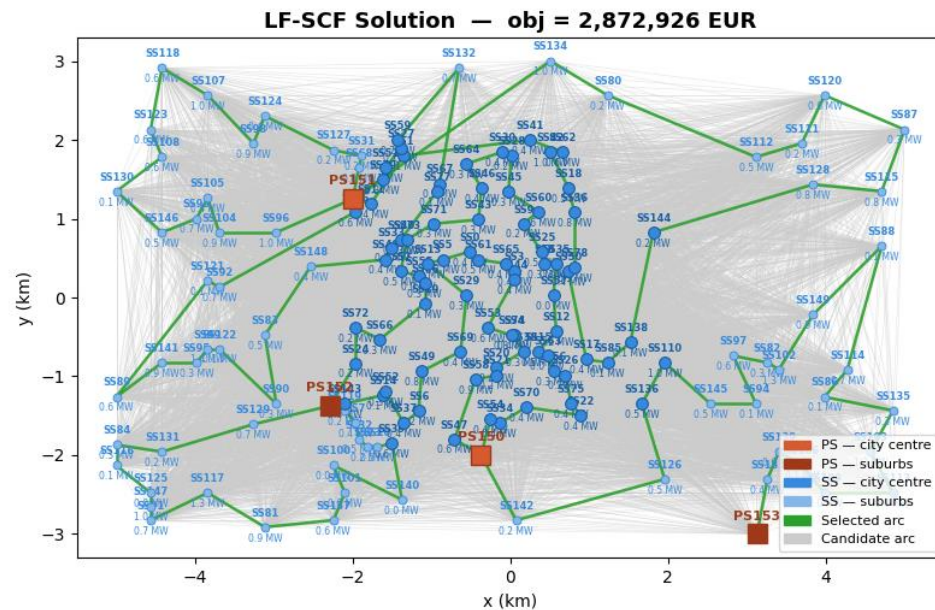


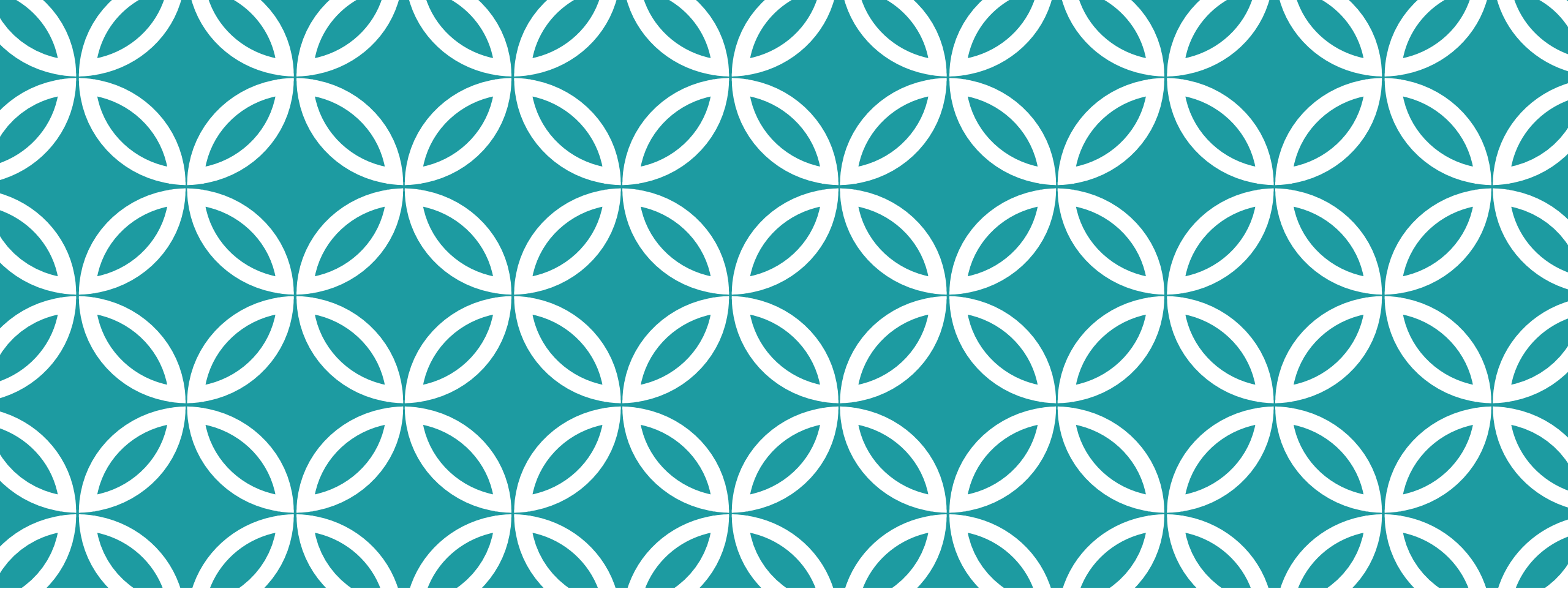
SOLUTION CASE154

#NODES	154
#ARCS	11781
Cable capacity	10 MW
Total demand	65.36 MW

Problem	Execution time	Gap	Root gap	Max execution time	Solution
LF-SCF	Limit reached	0.64%	10.39%	60m	2,872,926
LF-SE	Limit reached	3.12%	10.39%	60m	2,872,926
OL-SE	Limit reached	55.12%	57.51%	60m	6,064,629
OL-SE+CC	Limit reached	60.43%	62.73%	60m	6,908,385

SOLUTION CASE154



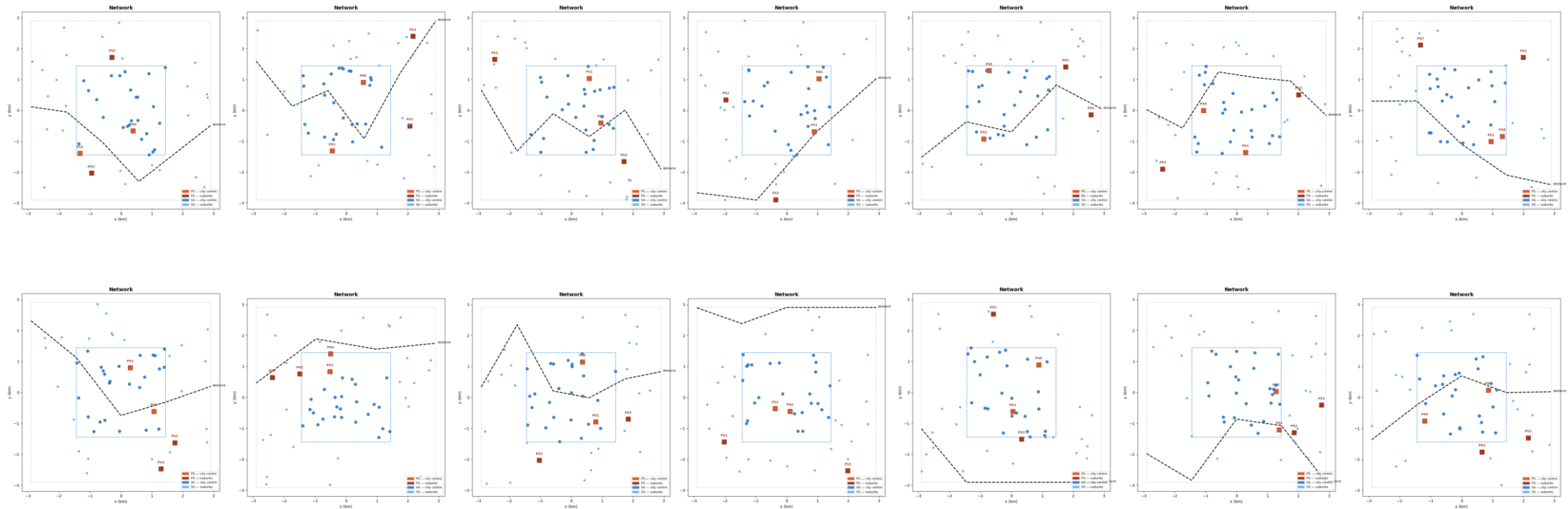


GENERATION

We can use a custom network generator to perform the scalability test

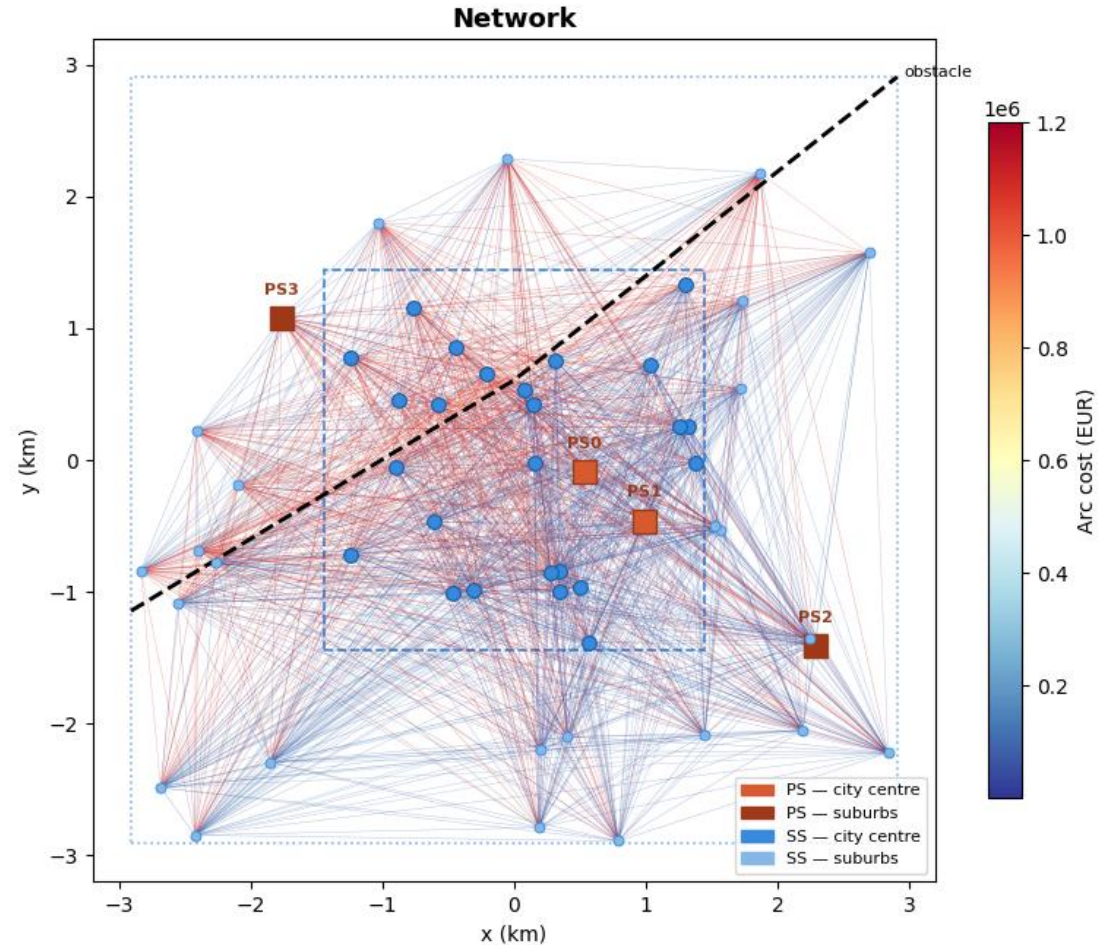
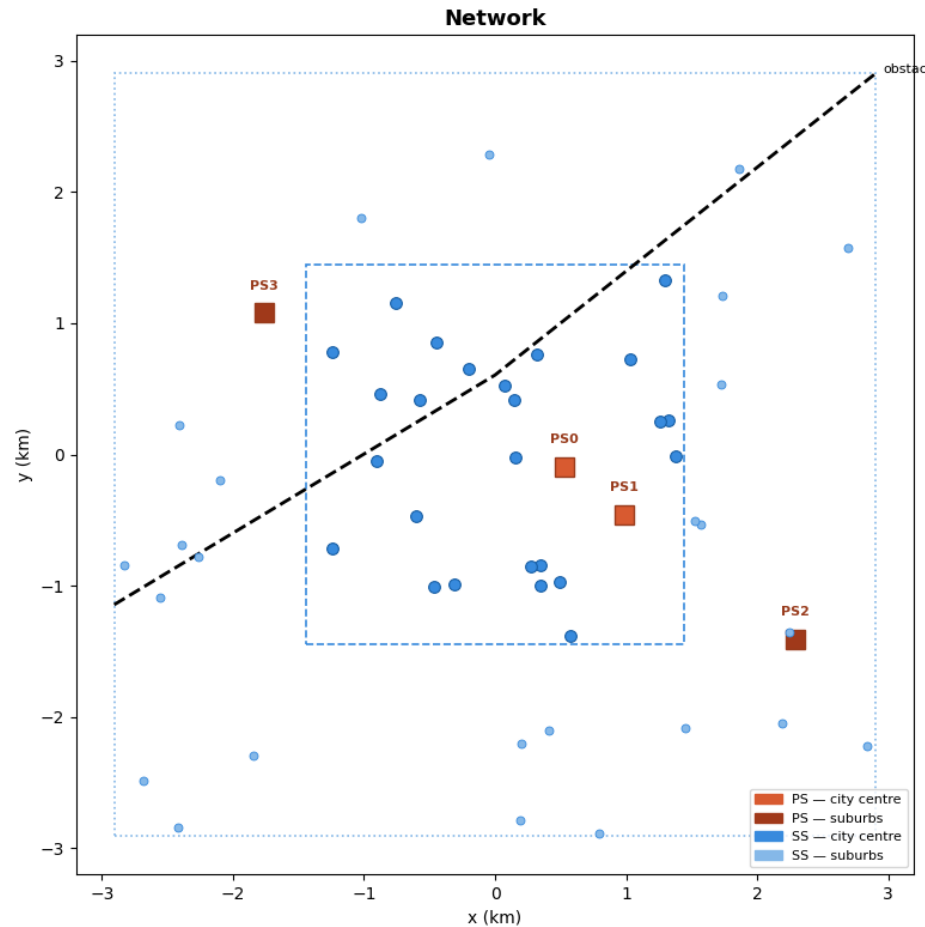
DATA SOURCE

Different variations of the same network from multiple SEEDs



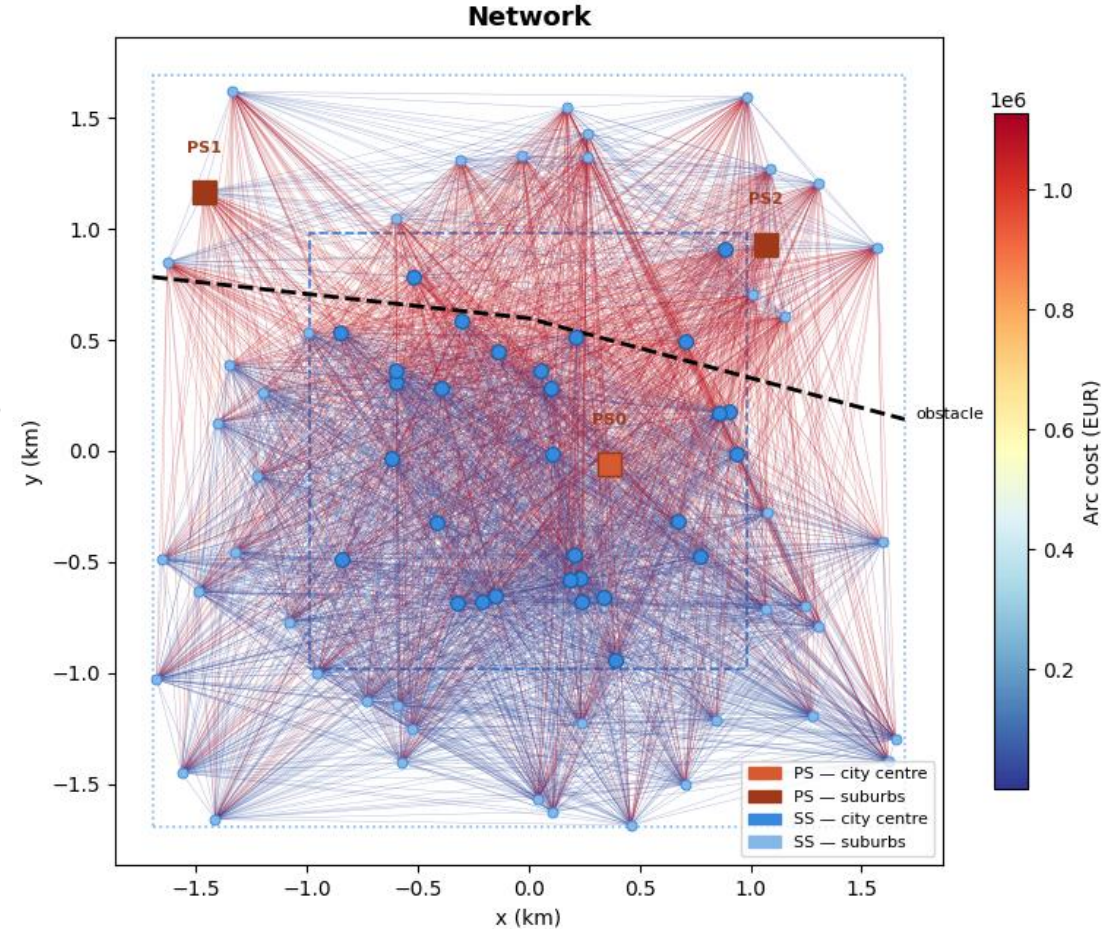
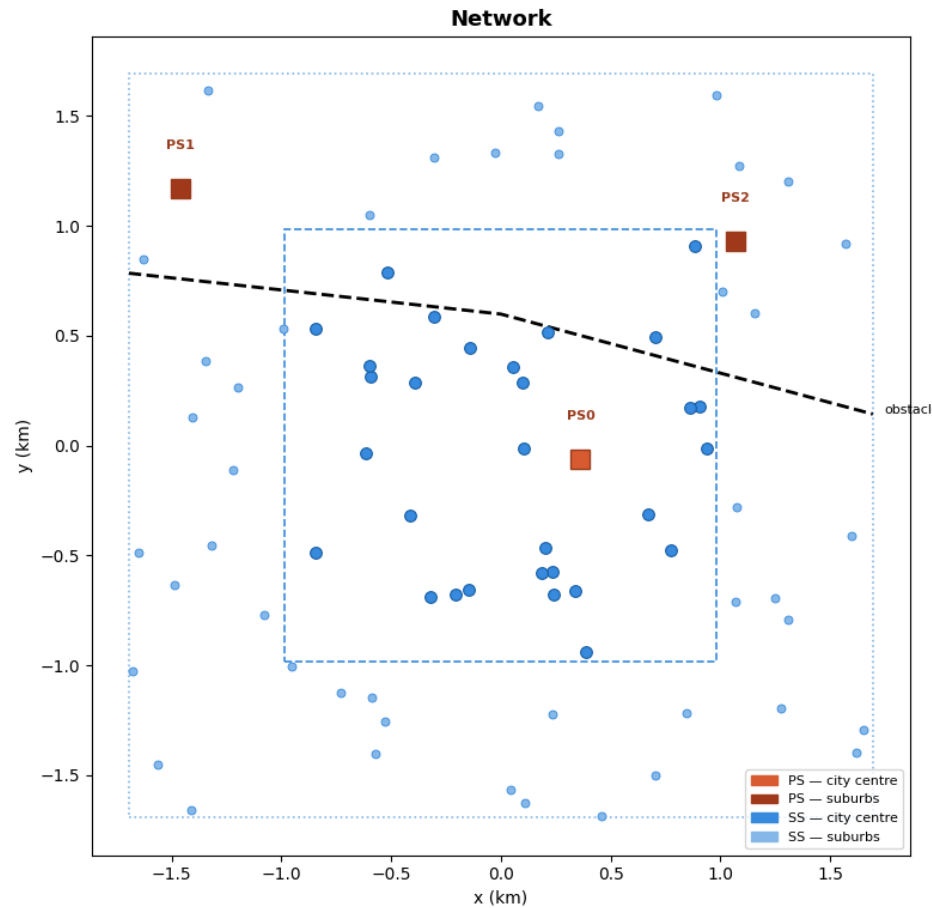
DATA SOURCE

Case study	PSs		SSs		Density (nodes/km2)	
	Center	Suburbs	Center	Suburbs	Center	Suburbs
Case 54	2	2	25	25	3,00	0,74



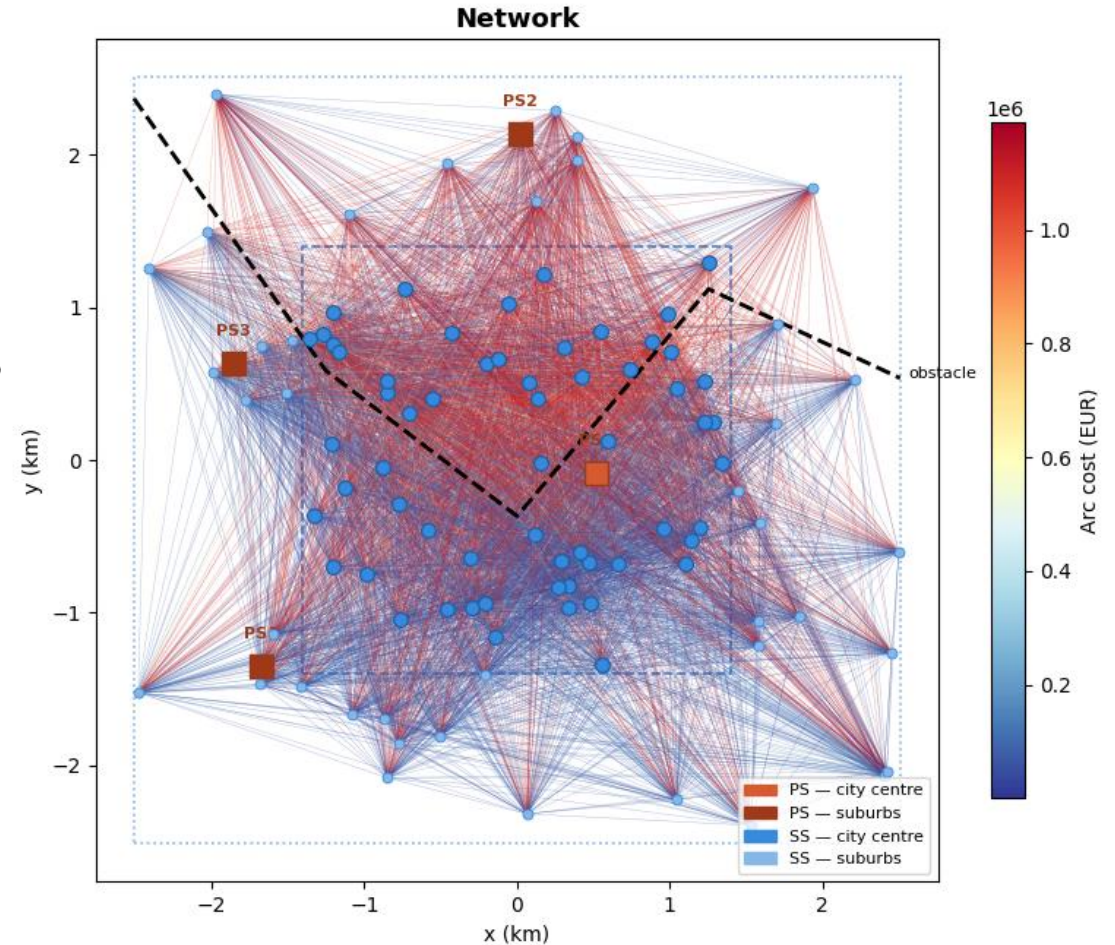
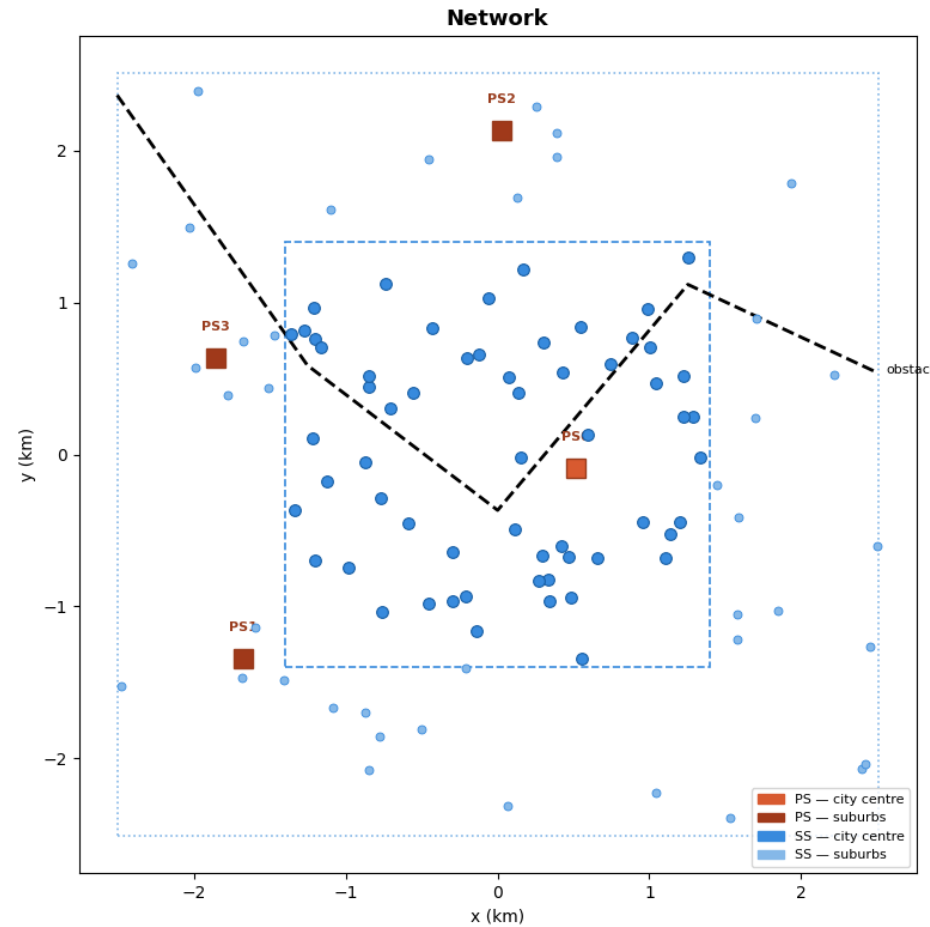
DATA SOURCE

Case study	PSs		SSs		Density (nodes/km ²)	
	Center	Suburbs	Center	Suburbs	Center	Suburbs
Case 78	1	2	30	45	7,75	3,92



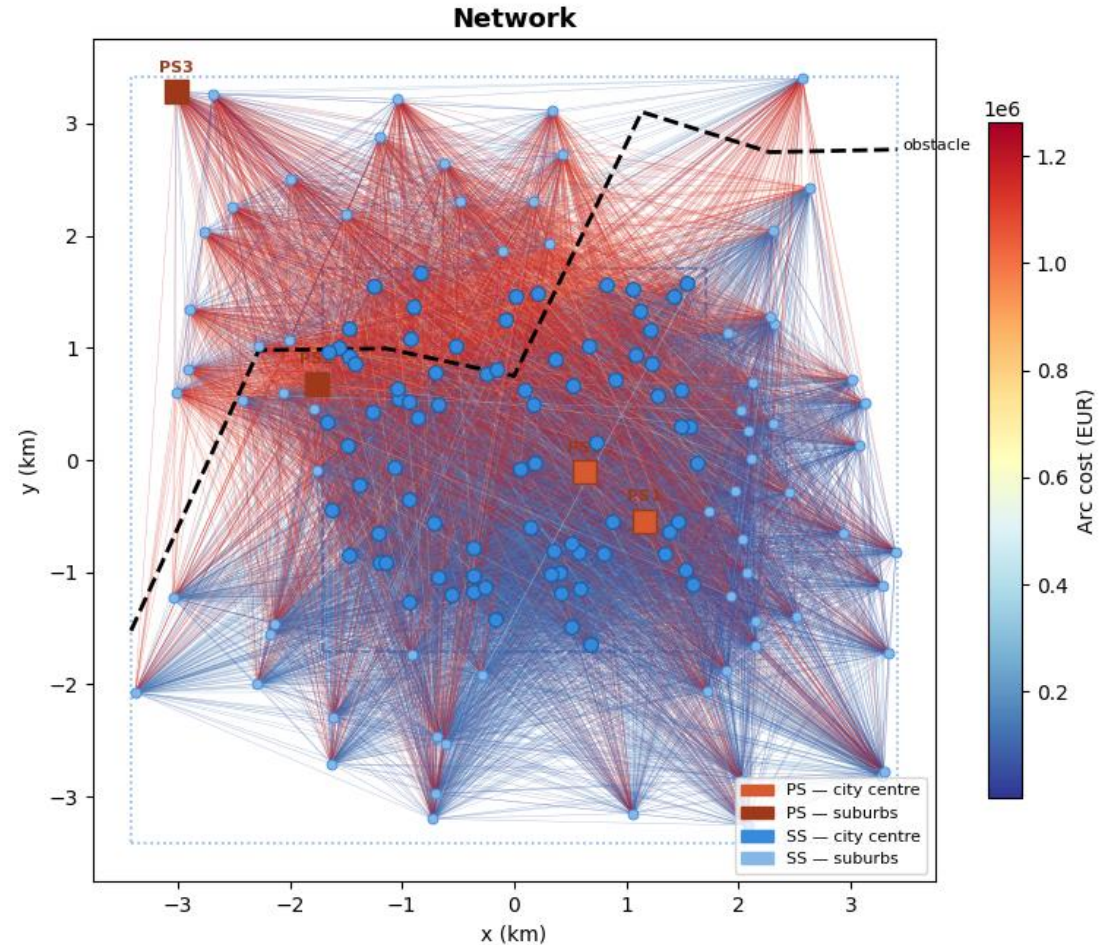
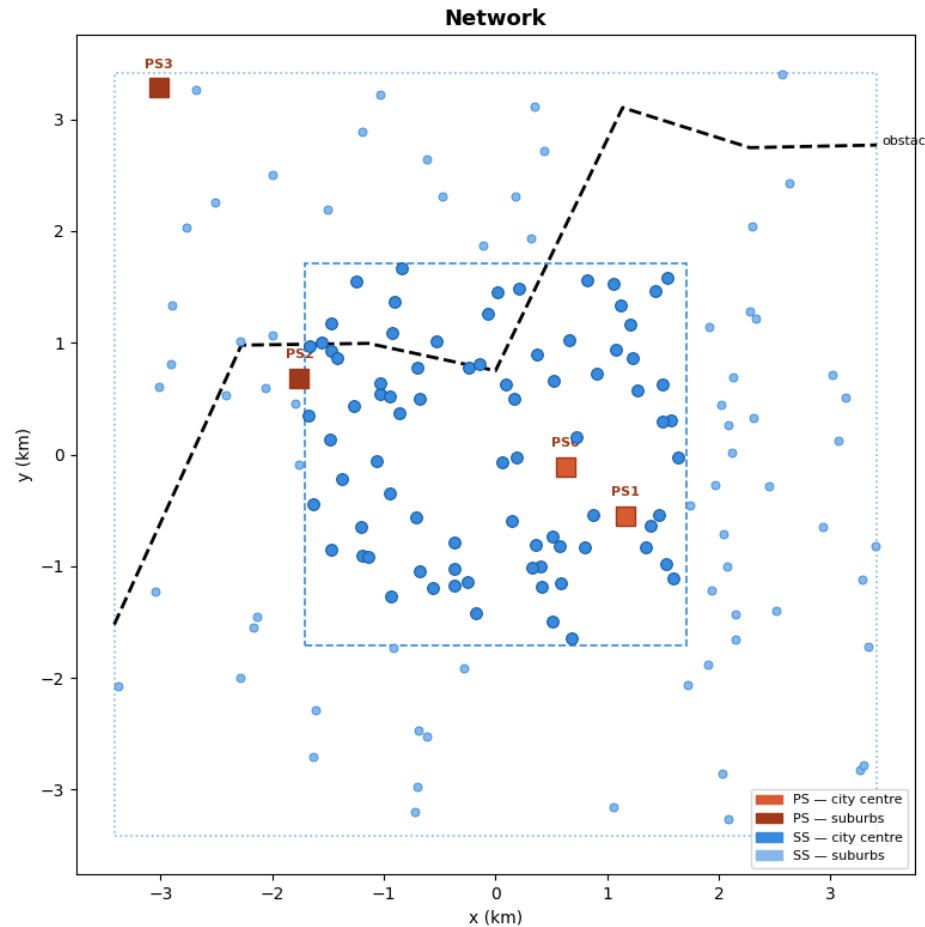
DATA SOURCE

Case study	PSs		SSs		Density (nodes/km2)	
	Center	Suburbs	Center	Suburbs	Center	Suburbs
Case 104	1	3	60	40	7,63	1,59



DATA SOURCE

Case study	PSs		SSs		Density (nodes/km2)	
	Center	Suburbs	Center	Suburbs	Center	Suburbs
Case 154	2	2	80	70	6,83	1,50



SCALABILITY TEST - METHODOLOGY

Network size schedule

- Networks grown incrementally: SS nodes from 3 to 200, PS nodes scaled from 1 to 4
- Each step generates a synthetic network via the **Generator** with a fixed base seed + step offset

Per-run time cap

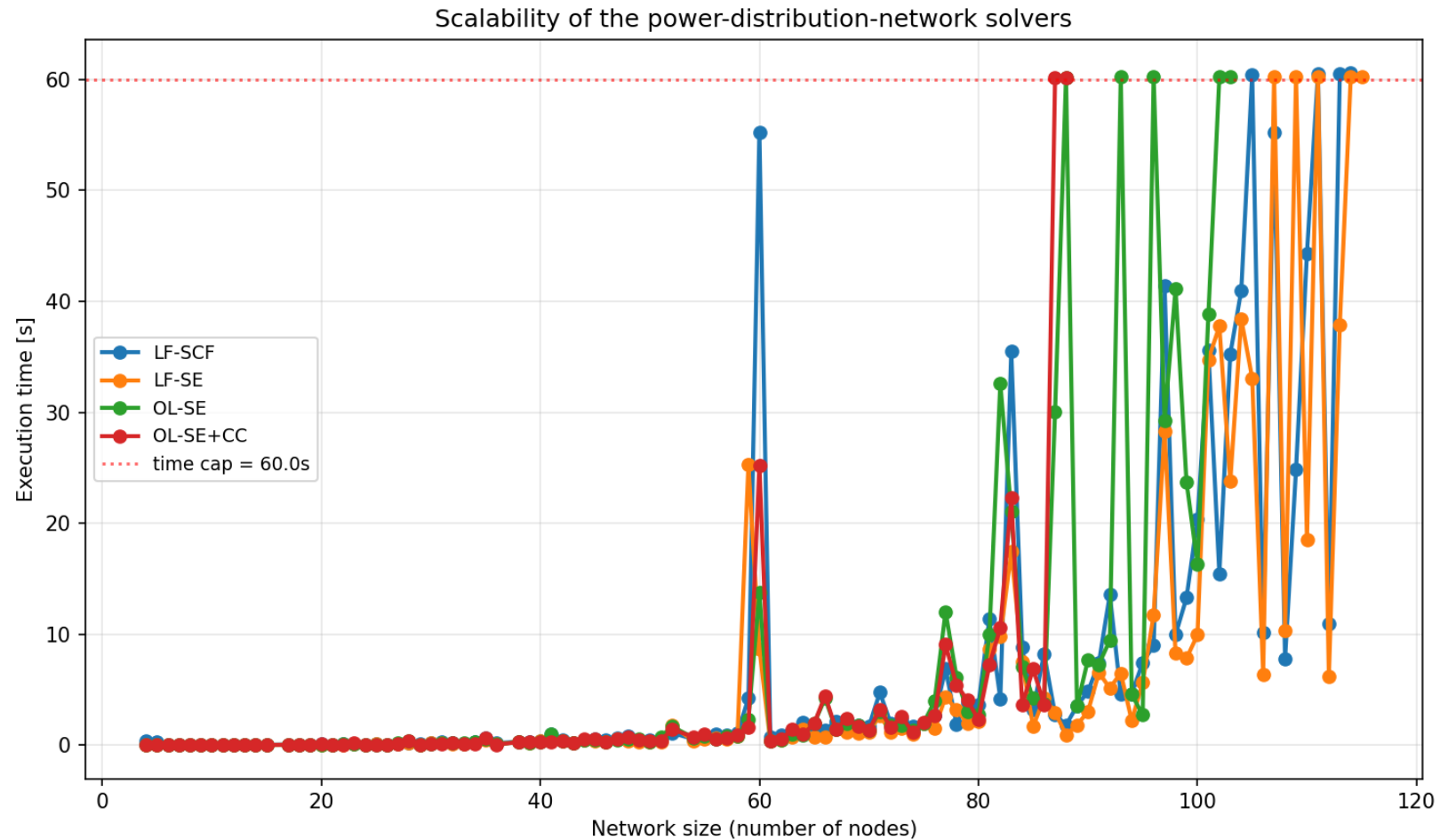
- Each solver is given a configurable time limit
- If a run exceeds the cap, **2 retries with different seeds** are attempted and the **median time** is recorded

Stopping criterion

- A solver is dropped after **2 consecutive overruns**; the test ends when all solvers are dropped



SCALABILITY TEST



SCALABILITY TEST - RESULTS SUMMARY

Solver	Runs completed	Max nodes	Mean time (s)*	Timeouts
LF-SCF	109	115	5.95	5 (5%)
LF-SE	108	114	4.55	4 (4%)
OL-SE	97	103	4.09	5 (5%)
OL-SE+CC	82	88	1.81	2 (2%)

* Computed on runs that completed within the 60 s cap.

Key takeaway: OL-SE+CC reaches fewest nodes (88) but is fastest on solvable instances (median 0.37 s, 2% timeouts). LF-SCF and LF-SE scale furthest (115/114 nodes) at the cost of higher variance.

SCALABILITY TEST - KEY OBSERVATIONS

Complexity is not determined solely by network size

- All solvers run in near-zero time for networks up to ~55 nodes
- Execution time spikes unpredictably at specific network sizes (e.g., ~60 nodes), suggesting the random configuration strongly influences difficulty

High variance across solvers for the same instance

- OL-SE+CC hits the 60s time cap earliest and most consistently for larger networks
- LF-SCF and LF-SE show similar behaviour but occasionally differ significantly on the same instance

Scalability is instance-dependent, not just size-dependent

- A solver may solve a 100-node network faster than a 90-node one, depending on the topology and constraint structure of the random configuration
- Benchmarking over multiple random seeds per size would be needed for reliable scalability estimates